RESEARCH EDITION v1.0

Q-Pathformer

Routed States

Classical Models and Quantum Circuit Simulation

Consolidated Research Findings



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Abstract

Q-Pathformer studies routed classical predictors and shallow two-qubit circuits emulated on a CPU, in preparation for a future physical-QPU comparison. We distinguish three completed experiments. In v0.6, a richer circuit readout improved checkerboard and spiral classification, but a freely trained routed trigonometric control remained stronger; exact conversion preserved selected functions and accelerated local CPU inference. In v0.9, fixed, learned and residual routing variants were compared on eight fresh populations per synthetic task with two paired optimization seeds. Twelve multiplicity-adjusted primary binary cross-entropy contrasts gave learned soft circuit routing five favorable, three unfavorable and four inconclusive comparisons; three favorable intervals exceeded the registered 0.01 practical-gain reference. The classical G121 control was stronger on checkerboard, spiral and disks, while the learned circuit was stronger on annulus. A separate Banknote check remained descriptive. In v0.9.5, 192 confirmation portfolios compared fixed allocation, capped successive halving and a registered Elastic Task Execution Layer (ETEL) suitability allocator over six fixed recipes under identical task-family CPU ceilings. None of eight adjusted native-BCE intervals established a practical ETEL gain. ETEL and halving matched observed test BCE exactly on three tasks; annulus slightly favored halving but remained inconclusive. Halving used less mean CPU on all four tasks. These results support task-dependent classical predictors, recoverable execution and a reproducible hardware interface. They do not establish a general ETEL efficiency gain, integrated architectural validation, language-generation ability or quantum advantage. Physical-QPU testing remains future work under a separate protocol.

Keywords: routed predictors; two-qubit CPU emulation; classical controls; training allocation; successive halving; reproducible machine learning.

The findings concern classical computation and CPU simulation. No physical quantum result is reported.

1 Introduction and contributions

Q-Pathformer is a research architecture for directing inputs through processing states and specialized parameter banks. Its broader proposal separates prioritization, shared representation and routing, and task-specific adaptation [1]. This paper evaluates bounded classical components of that direction: readout and deployment in v0.6 [2], routing and model choices in v0.9 [12], and training allocation in v0.9.5 [13]. These experiments address different questions and remain separate evidence strands.

QPATH-ARCH-001 records first publication on LinkedIn on 5 December 2024 [14]. Its retained PDF edition was created in March 2026. The October 2026 architecture proposal [1] and the companion Architecture and Experimental Specification [15] develop that conceptual lineage. The completed component experiments reported here do not retrospectively validate the entire original architecture.

The phrase “emulation of quantum states” refers here to numerical manipulation of a two-qubit statevector on an ordinary CPU. A four-component complex vector represents the amplitudes of $|00\rangle$, $|01\rangle$, $|10\rangle$ and $|11\rangle$; gate matrices evolve that vector, and expectation values become features for a learned classifier. Neither the router nor the classical control models require physical qubits. A compiled predictor can reproduce the selected circuit’s input-to-output function without reconstructing its quantum state. These are distinct implementations within a classical-computation study.

Publication v1.0 reports classical computation, including numerical two-qubit simulation. Labels v0.6, v0.9 and v0.9.5 identify separate studies rather than successive scores on one benchmark. Physical execution remains future work within the planned private v2.0 research programme; no outcome is assumed in advance.

The questions are whether richer readouts help a shallow circuit, whether learned routing helps against appropriate fixed and classical controls, whether selected predictors can be deployed efficiently on a CPU, and whether an adaptive allocator improves final prediction quality or resource use. The answer depends on the task, comparator and budget. We retain favorable, unfavorable and inconclusive outcomes, with the unit of replication and uncertainty stated for each study.

Technical overview of the completed studies

The v0.6 experiment compares six recipes on one confirmation population per task with five paired seeds and 15,000 updates: 120 endpoints. Its retained tables and figures occupy the original results sections below. The later v0.9 study expands population replication and tests nine synthetic model arms; the v0.9.5 study fixes a six-recipe candidate pool and compares three allocation policies. Model-level input routing and allocator-level training decisions are different mechanisms. Neither a replay PASS nor a higher score in a different experiment establishes an efficacy gain.

The earlier v0.1–v0.6 development history is retained in Appendix A. Later protocols and outcomes are identified separately. Versions use different tasks, controls, populations and objectives and are not independent replications of one common effect; no cross-version pooled score or improvement percentage is reported.

The contributions are controlled readout and routing comparisons, verified classical conversion, and an auditable allocation comparison that exposes a useful negative result for the registered ETEL policy. The Fourier representation and grid-based surrogate recovery have established prior art [5,6];

we do not claim to invent them. The full integrated Q-Pathformer architecture remains a separate hypothesis.

2 Related work and architectural context

2.1 Routed computation and the proposed states

The architecture proposal names Q-Sphere State 3 for priority, data quality and sampling; Cube State 2 for shared representation and routing; and Sphere State 1 for adapters, task heads, replay and controlled updates [1]. These are functional processing stages, not three physical quantum states. v0.6 uses a frozen nearest-centroid bank choice. v0.9 adds learned input gates, while v0.9.5 tests an allocator over whole training candidates. Neither follow-up establishes semantic prioritization, learned inter-stage transitions or continual retention.

Sparse mixture-of-experts systems use conditional activation to increase available capacity without activating every expert for every input [3]. The v0.6 fixed router is a narrower partition of two coordinates. The v0.9 soft variants evaluate all five experts and mix their probabilities; they are not sparse language-model gates. Bank count, gate and head parameters, function class and optimization geometry differ across controls, so routing cannot be inferred independently of these design choices from parameter count alone.

2.2 Encoded circuits and classical function classes

Data re-uploading repeatedly embeds a classical input between parameterized operations and provides a constructive approach to quantum classifiers [4]. The present core uploads each coordinate twice, within a two-layer, two-qubit circuit. It tests this particular finite recipe rather than a universal approximation claim.

Schuld, Sweke and Meyer connect data encoding to accessible Fourier frequencies [5]. Schreiber, Eisert and Meyer explicitly construct classical Fourier surrogates from a grid and compare directly optimized surrogate controls [6]. Their framework is the closest methodological precedent for the conversion studied here. Neural Fourier surrogates offer a different learned parameterization over corresponding frequency support [7]. These sources motivate suitable classical controls; they do not independently validate Q-Pathformer.

Our Trig25 control learns 25 real tensor-product coefficients freely. The depth-two circuit logits lie in the same span, but their trainable angles constrain the coefficients they can produce. Matching frequency support therefore does not imply equal effective capacity, identical optimization geometry or equal parameter count. Conversely, converting an already selected circuit logit into coefficients preserves its function; it does not retrain it as the more flexible Trig25 control.

Successive halving allocates progressively larger budgets to a shrinking set of candidates [24]. The v0.9.5 comparator uses a capped synchronous variant with six-to-three-to-one rungs, a 200-round survivor cap and permitted unused budget. Its measured outcomes belong to this implementation and candidate pool.

2.3 Evidential scope

The evidence comprises three internally controlled studies with different replication units. v0.6 has one population per task and five optimization seeds; v0.9 and v0.9.5 each use eight independent confirmation populations per synthetic task with two paired seeds nested within each population. v0.9 adds one descriptive Banknote split with five optimization seeds. The scope remains small

classification, CPU implementation and local simulator diagnostics. Scores do not establish LLM or image-model capability. Benchmark design and classical controls can materially change quantum-machine-learning conclusions [11].

3 Model and classical controls in v0.6

3.1 Routed circuit execution

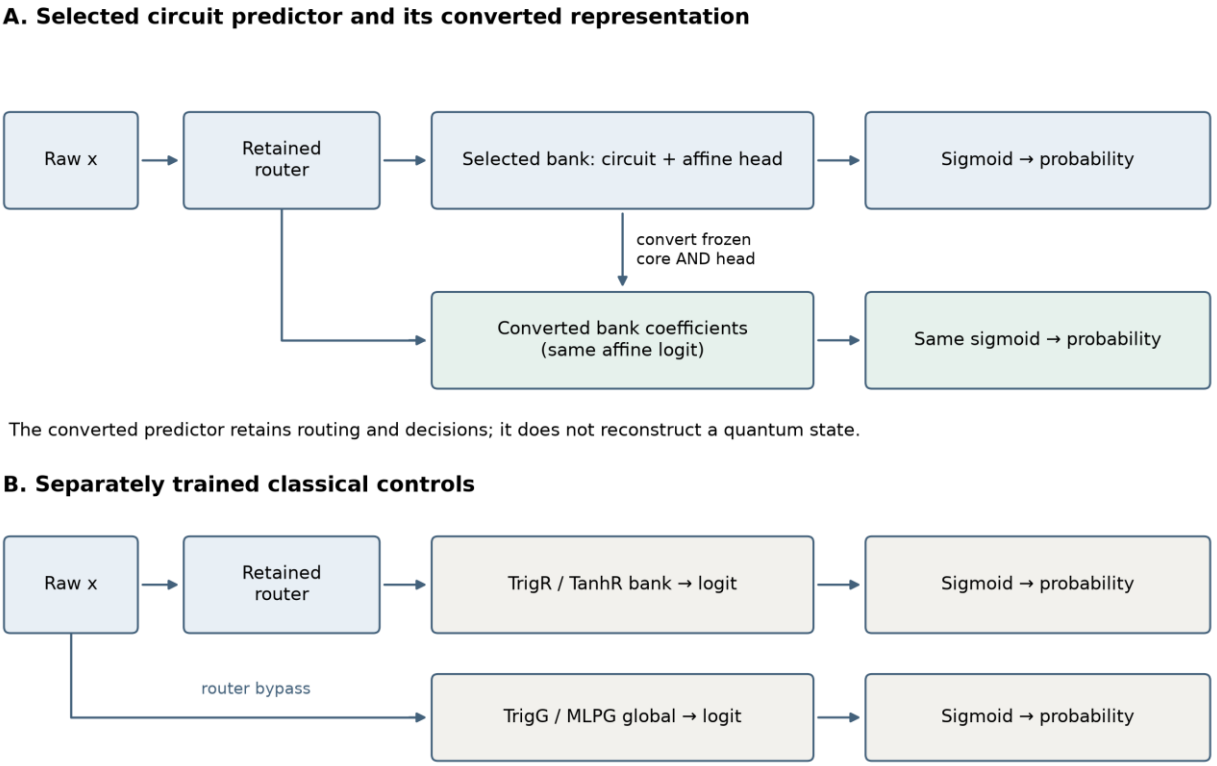


Figure 1. v0.6. Evaluated information paths, all executed classically. Conversion takes the frozen circuit and its learned affine head, preserving the router and sigmoid. TrigR and TanhR use routed banks; TrigG and MLPG bypass the router. Freely trained Trig coefficients are separate from converted circuit coefficients.

Let $x = (x_0, x_1)$ lie in $[-1, 1]^2$. One router per population is fitted using only the 4,800 training coordinates. It uses k-means++ initialization, 20 deterministic restarts and at most 500 Lloyd steps. The converged nonempty solution with minimum inertia is retained. Centroids are canonically ordered; the lowest index resolves equal distances. The same router serves all routed methods and training seeds in that population. Training inputs are canonically sorted. A Lloyd update stops only when assignments are unchanged and previous cluster counts are all positive; there is no centroid-motion tolerance. Empty centers are repaired from distinct largest-residual inputs. Eligible restarts are ordered by inertia, canonical centroid tuple and restart index. If none converges without empty clusters within 500 updates, fitting fails rather than accepting a fallback.

$$r(x) = arg\,min_k \|x - c_k\|^2, \quad k \in \{0, 1, 2, 3, 4\}. \qquad (1)$$

For each bank, the circuit begins in $|00\rangle$. In each of two layers, qubit q receives RY(πxq), trainable RY($\theta^{y}\ell q$), then trainable RZ($\theta^{z}\ell q$). A CNOT from qubit 0 to qubit 1 follows the two single-qubit sequences. The rightmost factor acts first in Equation 2. There are eight trainable core angles per bank and no legacy fixed final readout rotation.

$$A\ell q(xq) = RZ(\theta^{z}\ell q)\, RY(\theta^{y}\ell q)\, RY(\pi xq), \quad U\ell = CX_0 \to {}_1(A\ell 0 \otimes A\ell 1).$$

$$|\psi\theta(x)\rangle = U_1(x)\, U_0(x)\, |00\rangle. \qquad (2)$$

3.2 Readouts and classical controls

An observable O supplies the moment $mO(x) = \langle \psi\theta(x)|O|\psi\theta(x)\rangle$. Scalar uses $X_o = X \otimes I$. Multi uses the ordered vector $(X_o, X_1, Z_o, Z_1, Z_oZ_1)$. Both apply a trainable affine head and sigmoid. All decisions use $p \geq 0.5$. The core architecture and initialization, raw inputs, routing policy and initial predictions are paired; the learned angles can diverge. Head dimensionality and available model capacity differ.

$$zScalar(x) = w\, mX_0(x) + b, \quad zMulti(x) = w^{\mathrm{T}} m(x) + b, \quad p(x) = \sigma(z(x)). \quad (3)$$

Model	Predictor in each active bank	Banks	Parameters
Scalar	Two-qubit core; X_o affine head	5	50
Multi	Same core; five-moment affine head	5	70
TanhR	Two inputs; 8-unit tanh hidden layer	5	165
TrigR	25-coefficient trigonometric logit	5	125
TrigG	Same basis; global logit	1	25
MLPG	Two inputs; 32-unit ReLU hidden layer	1	129

Table 1. v0.6. Allocated trainable predictor parameters. The ten fitted centroid coordinates are separate. These recipes are not parameter matched. Global controls bypass the router.

For TrigR and TrigG, define $\phi(t) = [1, \cos(\pi t), \sin(\pi t), \cos(2\pi t), \sin(2\pi t)]$. The logit is the tensor product in Equation 4, with the constant coefficient providing the bias. TanhR and MLPG receive the same two raw coordinates, not circuit moments or precomputed trigonometric inputs. Each has one hidden layer and an affine output logit.

$$zTrig(x) = \Sigma_{a=0}{}^{4}\, \Sigma_{\beta=0}{}^{4}\, C_{a\beta}\, \varphi_a(x_0)\, \varphi_\beta(x_1). \quad (4)$$

Circuit banks use PCG64 SeedSequence([seed,1]) and uniform [−0.1,0.1) core angles; Scalar and Multi share the initial draw. X_o head weight starts at 2; remaining weights and biases start at zero. For width h (8 for TanhR, 32 for MLPG), input weights are uniform on $[-\sqrt{(6/(h+2))}, \sqrt{(6/(h+2))})$ and output weights on $[-\sqrt{(6/(h+1))}, \sqrt{(6/(h+1))})$; biases are zero. The MLP generator uses SeedSequence([seed,4,h]) with bank-by-bank draws, without an activation-specific gain. Trig coefficients start at zero; their seed variation reflects record order. All predictor parameters train while routing stays fixed.

There is no hash-routed training arm in v0.6. Hash metadata retained by a shared data-preparation helper is provenance, not an input feature or a seventh predictor. This distinction matters when relating the component study to the earlier Hash18 text experiment.

4 Experimental design and analysis

Sections 4.1–4.4 retain the v0.6 design and uncertainty. The follow-up designs in Sections 4.5–4.6 use separate populations, protocols and statistical families. No later result is added to the v0.6 sample size.

4.1 Populations and experimental separation

Inputs are generated uniformly on $[-1, 1]^2$. Each population contains 8,000 records, split into 4,800 training, 1,600 validation and 1,600 test records using observed-label stratification. The split preserves natural prevalence; neither rare classes nor routed banks are balanced by resampling. Latent labels and noise flags are excluded from training and checkpoint selection.

Task	Binary label construction	Purpose
Checkerboard	$1[\sin(3\pi x_0 + \pi/7)\sin(3\pi x_1 + \pi/5) > 0]$	Repeated, sharp boundaries
Spiral	$1[\sin(2\,\mathrm{atan2}(x_1,x_0) + 4\pi\sqrt{(x_0^2+x_1^2)}) > 0]$	Curved oscillating boundary
Noisy annulus	Latent $1[0.25 < x_0^2+x_1^2 < 0.85]$; independent 10% label flip	Geometry under irreducible label noise
Rare disks	Union of three radius-0.22 disks centered at (−0.55,−0.45), (0.50,−0.35), (0,0.55)	Low positive-class prevalence

Table 2. v0.6. Registered generators. Indicator $1[\cdot]$ is one when its condition holds. The annulus primary target is the observed noisy label. Boundary definitions, random seeds and exact arrays are retained in the evidence package.

Separate engineering, development and confirmation populations are generated for each family: 12 populations and 96,000 unique coordinate pairs. Freshness checks exclude 168,000 historical coordinate pairs drawn from 65 earlier NPZ files, with signed-zero normalization and exact canonical coordinate comparisons. The four confirmation populations are independent data realizations of their registered families, not additional task designs selected after viewing test results.

Engineering contains 24 short endpoints, one per recipe and task. Stage A contains 48 full endpoints using seeds 1101–1102. Stage B contains 120 full endpoints using seeds 1201–1205. Every full endpoint completes 200 rounds and 15,000 optimizer updates. All six arms proceed regardless of development rank. The confirmation test outputs remain inaccessible to model selection until all required endpoints are completed, committed and independently audited, followed by a hash-bound evaluation release.

The resulting design supports comparisons among the registered recipes on each fixed fresh population. Five training seeds do not represent five independently sampled datasets. We therefore report each task separately and distinguish seed variability from test-record resampling. Earlier experiments are summarized as historical motivation in Appendix A; their different tasks and designs are not pooled into the confirmation estimates.

4.2 Training and checkpoint selection

Each round shuffles all 4,800 training rows and uses 75 minibatches of 64. Record orders are identical across methods for each population and seed. All methods minimize stable native-logit binary cross-entropy (BCE), using Adam [8] with learning rate 0.01, $\beta_1 = 0.9$, $\beta_2 = 0.999$ and $\varepsilon = 10^{-8}$, bias correction, no weight decay, no learning-rate schedule and no gradient clipping.

$$L(z,y) = mean[max(z,0) - yz + log(1 + exp(-|z|))]. \quad (5)$$

Circuit expectation derivatives use the parameter-shift identity with $\pm\pi/2$ shifts for each of eight angles [10]. The affine head and loss use the analytic chain rule; MLP and Trig gradients use manual

analytic derivatives. A circuit training record requires one ordinary and sixteen shifted statevector evolutions. Bank gradients are normalized by the total minibatch size. Adam operates on the full packed parameter vector: an absent bank has zero fresh gradient, but existing momentum can still change its parameters. This is not strict sparse optimizer updating.

$$\partial m/\partial \theta_{\mathrm{j}} = [m(\theta_{\mathrm{j}} + \pi/2) - m(\theta_{\mathrm{j}} - \pi/2)] / 2. \qquad (5a)$$

Validation runs after every round. The selected checkpoint is the earliest exact minimum of full-validation common BCE, calculated after clipping probabilities to $[10^{-7}, 1-10^{-7}]$. All rounds complete regardless of selection or F1 milestones. Native logits and unbounded native losses are retained because the bounded common score can conceal extreme wrong predictions. No test threshold tuning, post-test calibration or arm reselection is performed.

4.3 Estimands and uncertainty

The primary outcome is common natural-log BCE, lower being better. Four prespecified contrasts per task form one family of 16: Multi minus Scalar, Scalar minus TrigR, Multi minus TrigR, and TrigR minus TrigG. Paired differences across five seeds use two-sided t intervals with four degrees of freedom and Bonferroni confidence 99.6875%. A BCE magnitude of 0.01 is a registered practical reference, not a universal application threshold.

“Registered” in this paper means an internally frozen protocol, not public preregistration. The analysis protocol was bound at Stage A approval (19 October 2026, 05:14:11 UTC); Stage B was frozen at 05:48:39 UTC and launched at 05:48:43 UTC. Confirmation evaluation was released only after endpoint audit, at 06:37:52 UTC. Exact protocol, freeze and access-event hashes are in CLASSICAL_REVISION_EVIDENCE.json.

A separate ordinary, unstratified paired-record bootstrap uses 10,000 resamples of the 1,600 test records. Identical record counts are applied to all methods and five fixed checkpoints; metrics are computed per checkpoint and averaged. Nominal Bonferroni-adjusted percentile intervals condition on these fits, rather than sampling new datasets, routers or training. Bootstrap draws use PCG64 seeds 298601–298604 in fixed task order; endpoints use NumPy linear quantiles. At 99.6875% confidence, each tail has probability 0.0015625, about 15.6 of the 10,000 replicates. Extreme endpoints can therefore have Monte Carlo variability; multiplicity adjustment does not remove bootstrap coverage approximation. The original registered analysis is retained without a new larger-resample analysis.

Macro-F1 averages class-wise F1 at p ≥ 0.5 and is distinct from accuracy. F1, raw Brier score, 10-bin probability ECE, class precision/recall, average precision and native loss are secondary descriptive measures. Validation steps to 90%, 95% and 97% macro-F1 require three consecutive checks; the third confirms attainment. Nonreachers remain censored at 15,000 updates. These descriptive milestones never stop training.

4.4 Hardware and computational accounting

Item	Recorded study environment
Processor	AMD Ryzen AI 9 HX 370 w/ Radeon 890M; 12 physical cores, 24 logical processors
Memory	67,772,469,248 bytes reported by the OS; approximately 63.12 GiB
Operating system	Windows 11 Pro; version 10.0.26300; build 26300

Item	Recorded study environment
Numerical runtime	Python 3.12.14, NumPy 2.5.4, SciPy 1.18.1; float64 parameters and complex128 statevectors
Local circuit tools	Qiskit 2.6.0; Qiskit Aer 0.17.2; separate simulator environment with qiskit-ibm-runtime 0.50.0
Training execution	CPU only; Stage A: two workers; Stage B: five workers; one BLAS thread requested per process
Inference execution	Isolated serial CPU benchmark; one process and one BLAS thread; batch sizes 1, 32 and 1,024
Accelerators	No GPU model execution or physical quantum-processor job in this study

Table 3. Historical execution environment from sealed runtime records [2]. Installed graphics in the CPU name do not indicate GPU use. This is the machine used for the experiments, not the document-authoring environment.

Stage A's launcher elapsed time was 1,930.443 s (32.17 min), and Stage B's was 2,760.768 s (46.01 min). These are wall-clock phase durations under different worker concurrency. They exclude earlier engineering, subsequent audit and evaluation, simulator and deployment work, report preparation and sealing. They are not per-model latency or additive CPU-core time.

Each full endpoint performs 960,000 training-record visits. For either circuit recipe this corresponds to 16,320,000 statevector evolutions for training gradients alone. This count excludes validation, checkpoint replay, diagnostics and local readiness work. Moment extraction from a statevector is exact numerical computation; logical measurement-setting equivalents are accounting quantities, not training shots on a device.

Training and validation route assignments are precomputed and charged. There is no persistent circuit-moment or trigonometric-design cache. The inference benchmark includes fresh raw-input routing, feature construction and sigmoid evaluation; checkpoint loading and coefficient construction are measured separately. The future hardware cost of state preparation, grouped measurements, classical postprocessing, queueing and mitigation cannot be inferred from these CPU times.

The retained records do not establish the BLAS vendor/build, measured CPU clock, energy consumption, GPU utilization traces or peak process-memory telemetry. The inventory MaxClockSpeed field is not treated as a measured experimental clock. The paper consequently reports actual observed latency and stored numerical bytes, without claims about joules, FLOPs, accelerator speedup or hardware resource superiority.

4.5 Routing and model comparison in v0.9

v0.9 compared circuit arms C_H, C_S, C_L and C_R; analogous freely trained trigonometric arms T_H, T_S, T_L and T_R; and the global classical G121 control [12]. H is hard geometric routing, S fixed soft routing, L learned soft routing, and R fixed soft routing with a 12-parameter output residual. The circuit arms have 70, 70, 82 and 82 trainable parameters; the trigonometric arms have 125, 125, 137 and 137; G121 has 121. C_S and C_L emulate two qubits on a CPU. G121 is a classical baseline, not a circuit or QPU result.

Five centroids were fitted from training data with 20 restarts and a 500-iteration cap; the soft temperature was fixed from training-coordinate distance gaps, without labels. Soft variants mix expert probabilities, not logits, and evaluate all five experts. The learned gate uses four affine corrections with the fifth fixed to zero; the residual uses 12 degree-two-to-four monomials. Paired S/L/R arms share expert initialization and sampling. Gate/residual parameters start at zero, remain frozen for 750 updates and then use local Adam step 1. Expert learning rate is 0.01 and gate/residual rate 0.001. Float 64 Adam uses 0.9/0.999 moments and 1e−8 epsilon, batch 64 sampling with replacement, 75 updates per round and 200 rounds. Settings were fixed rather than separately optimized for each family.

G121 uses $b_5(t) = (1, \cos(\pi t), \sin(\pi t), \ldots, \cos(5\pi t), \sin(5\pi t))$ and $\Phi(x) = b_5(x_0) \otimes b_5(x_1)$. Its prediction is $\sigma(w^T\Phi(x))$, with 121 freely trained real coefficients initially zero; the constant×constant coefficient is the intercept, with no extra bias. Synthetic inputs retain their two coordinates in $[-1, 1]^2$. Banknote inputs are standardized using training means and population SDs, projected onto two training-fitted PCA axes, standardized again on training PCs, then bounded coordinate-wise by $(2/\pi)$ arctan. The basis uses these final coordinates, multiplied by π inside the trigonometric functions. [12, quantum/models.py, trig_features and Model._expert; quantum/routers.py, fit_preprocessing and preprocess]

For fixed-soft routing, $d_j(x) = \|x - \mu_j\|^2$ and $q_j(x) = \exp(-d_j/\tau)/\Sigma_l \exp(-d_l/\tau)$, over five frozen centroids. The training-only rule is $\tau = \max(0.001, \text{median}(d_{(2)} - d_{(1)}))$, where $d_{(1)}$ and $d_{(2)}$ are the nearest and second-nearest squared distances for each training input. Centroids minimize eligible k-means inertia; temperature is a geometric statistic, not a label-loss fit. Learned-soft routing adds $a_j^T(x_0, x_1, 1)$ to $-d_j/\tau$ for the first four experts before softmax, fixing the fifth correction to zero: four affine triples give 12 additional parameters. [12, quantum/routers.py, fit_route_policy; quantum/models.py, Model.gate]

The residual arm keeps the fixed-soft gate and forms $m(x) = \Sigma_j q_j(x)\sigma(z_j(x))$. It predicts $\sigma(\text{logit}(m(x)) + \beta^T r(x))$. The 12 entries of r are x_0^2, x_0x_1, x_1^2; x_0^3, $x_0^2x_1$, $x_0x_1^2$, x_1^3; and x_0^4, $x_0^3x_1$, $x_0^2x_1^2$, $x_0x_1^3$, x_1^4, each divided by max(1, its maximum absolute training value). There is no additional constant or linear residual. Thus the correction enters after expert-probability mixing and before the final sigmoid. Every soft arm evaluates all five experts. [12, quantum/models.py, RESIDUAL_POWERS and Model._forward]

Forty fresh synthetic populations comprise four engineering, four development and 32 confirmation populations, with 4,800 training, 1,600 validation and 1,600 hidden-test records each. Eight confirmation populations per task and two paired seeds yield 576 synthetic endpoints across nine arms. Together with 60 Banknote endpoints, the confirmation set contains 636 fits. Engineering uses 48 short fits, development 72 full fits, and four clean retrainings complete 760 fits. Preprocessing and router fitting use training data only; validation selects endpoints before hidden evaluation.

The primary v0.9 endpoint is common BCE after clipping probabilities to [1e−7, 1−1e−7]. The 12 primary comparisons are C_L minus C_S, C_R and G121 within each of four tasks. Paired seed differences are averaged within population; eight population means support t intervals with df7 and Bonferroni confidence 99.583333%. An interval wholly below −0.01 establishes the registered practical-gain reference. Twelve secondary contrasts use nominal 95% intervals and remain exploratory; they are not a second corrected primary claim.

The Banknote Authentication dataset contains 1,372 records (UCI DOI 10.24432/C55P57, CC BY 4.0) [23]. Identical four-feature rows were grouped before one fixed 60/20/20 split. Standardization, PCA and bounded transforms were fitted on training data. Most arms use two PCA coordinates; MLP4

uses all four inputs and also differs in capacity. Five optimization seeds on this one split are descriptive, not five independent datasets. Absent specimen/session identifiers limit independence claims beyond the duplicate grouping.

4.6 Training allocation in v0.9.5

v0.9.5 fixes six recipes: C_S, C_L and G121 at learning-rate multipliers 0.5 and 1.0. It compares fixed allocation, capped synchronous successive halving [24] and the frozen ETEL suitability rule under identical task-family native-CPU ceilings [13]. It uses another 40 fresh populations with the same 4,800/1,600/1,600 split sizes, including eight confirmation populations per task and two paired seeds. Historical suitability guidance uses only 24 retained v0.9 development training/validation trajectories, never historical test outcomes. Operational cost calibration comes from v0.9.5 engineering and the reviewed amendment using the complete original development cohort, including failed attempts. Training-only preprocessing and the 200-round maximum remain fixed.

The execution component is Q-Path Multiplier - Elastic Task Execution Layer (ETEL) v0.1. Study v0.9.5 identifies this allocation experiment; research edition v1.0 identifies the present collection. Incorporated source 0.2.0 is a historical source identifier, not another name for the component release. [13]

Let B denote the task-family portfolio native-CPU ceiling, including charged overhead. B is 0.80 times the engineering cost of six full fits: 0.70 for training and a 0.10 overhead allowance, with all charges inside the ceiling. Every policy first gives all six candidates 60 of their maximum 200 rounds. The 30% point refers to each candidate's round horizon, not accuracy, elapsed time or portfolio CPU; it is a suitability review, not automatic termination. Fixed allocation advances candidates evenly. Halving retains six, then three at round 60, then one at round 120, using rung validation BCE and recipe-ID ties. Its survivor remains capped at 200, so unused budget is allowed.

ETEL estimates nonnegative improvement slope g over the preceding 20 rounds. For an affordable block of k rounds its score is max(0, L_best − L_i + min(0.02, k*g)) / (k*c_i + h_i), where L_i is current native validation BCE, L_best the best eligible grid value, c_i mean observed round CPU in the window and h_i the switch-cost estimate. Thus 0.02 caps the projected BCE improvement before division by cost. Normally k is ten; a one-round tail is considered when a full block is unaffordable. Displacing a feasible current candidate requires a positive-score challenger with at least a 10% score advantage and a forecast at least 0.001 BCE below L_best. All-zero scores fall back to lowest current BCE, with candidate-ID ties.

In the frozen controller, $g = \max(0, L_i(t-20) - L_i(t))/20$ for the completed 20-round window; if the latest retained older observation has a larger round gap, that actual gap is used. c_i averages measured round CPU over the same interval. L_best is the minimum available validation value on completed 10-round checkpoints across all candidates; an eliminated candidate's retained grid value remains eligible. For scoring, h_i is zero for the current candidate, otherwise its mean observed switch cost, falling back to the accepted cost calibration when no switch sample exists. Admission uses the maximum observed switch cost, not that scoring mean. The reviewed cost amendment retained failed charges and the original B values. [13, execution/controller.py, score, reservation and record_round; governance/ROOT_COST_AMENDMENT_ACCEPTANCE_01.json]

ETEL protects 10% of B for ten-round probes after the common floor. Every fourth post-floor decision offers a probe to an eligible paused candidate with least progress, with preseeded tie order; an affordable probe may occur earlier when normal work cannot fit without the reserve. The reserve is released only when no eligible ten-round probe fits. Probes stay inside B and cannot guarantee late-

improver discovery. All block admission uses the frozen 1.25 reservation multiplier and switch/closing terms. Expert rates are 0.005/0.01 and gate rates 0.0005/0.001 after 750 warm-up updates. Python 3.12.14, NumPy 2.5.4 and SciPy 1.18.1 use one OpenBLAS thread and at most five isolated portfolio workers, each advancing candidate states serially.

Original development ended with 20 successful and 4 failed jobs. A reviewed cost-only recalibration retained all failed charges and unchanged task ceilings, recipes, controller rules and reservation multiplier. Replacement 24 portfolios used fresh initialization on the same four development populations; they created no new confirmation populations. Confirmation contains 192 portfolios (3 policies ×4 tasks ×8 populations ×2 seeds), plus 192 separate full reference fits and 4 preselected clean prescribed-action replays. Hidden evaluation followed committed endpoints, validation selections and independent audits; 384 selected/reference models were officially scored.

Selection minimizes native validation BCE on the 10-round grid plus final committed endpoints, with earliest-checkpoint and recipe-ID ties; committed candidates remain eligible after elimination. Eight primary ETEL-minus-fixed/halving contrasts use native stable, unclipped BCE, paired seed averaging within population, n = 8 and df7. Bonferroni confidence is 99.375%. The practical-gain reference is −0.01 and the noninferiority reference +0.01; this design does not establish formal equivalence. Secondary metrics are descriptive. Because v0.9 clips common BCE and uses other populations and objectives, cross-version BCE differences are not an intervention effect.

5 Predictive results

The initial predictive-result tables and Sections 5.1–5.3 report v0.6. Sections 5.4 and 5.5 report v0.9 and v0.9.5 separately, with their own populations, metrics and uncertainty.

Primary common BCE (lower is better)

Model	Checkerboard	Spiral	Noisy annulus	Rare disks
Scalar	0.5780 ± 0.0120	0.5007 ± 0.0029	0.4106 ± 0.0017	0.0135 ± 0.0007
Multi	0.5216 ± 0.0031	0.4253 ± 0.0084	0.4091 ± 0.0024	0.0146 ± 0.0006
TanhR	0.6511 ± 0.0066	0.5682 ± 0.0227	0.4322 ± 0.0095	0.0523 ± 0.0296
TrigR	0.3050 ± 0.0014	0.2483 ± 0.0009	0.4032 ± 0.0010	0.0149 ± 0.0002
TrigG	0.6965 ± 0.0003	0.6679 ± 0.0004	0.4597 ± 0.0005	0.0167 ± 0.0002
MLPG	0.5667 ± 0.0343	0.4695 ± 0.0415	0.4184 ± 0.0031	0.0523 ± 0.0124

Table 4a. v0.6. Primary test BCE, mean ± sample SD across five optimization seeds on one fixed split/router per task.

Secondary macro-F1 (%) (higher is better)

Model	Checkerboard	Spiral	Noisy annulus	Rare disks
Scalar	73.03 ± 1.52	77.60 ± 0.60	86.20 ± 0.23	99.12 ± 0.20
Multi	78.55 ± 0.48	84.71 ± 0.58	86.92 ± 0.32	98.85 ± 0.20
TanhR	60.02 ± 1.00	66.05 ± 2.00	83.07 ± 1.54	94.17 ± 4.72
TrigR	89.57 ± 0.39	92.05 ± 0.32	86.81 ± 0.16	98.67 ± 0.12
TrigG	48.89 ± 1.15	57.57 ± 0.35	82.14 ± 0.26	99.30 ± 0.08
MLPG	69.55 ± 4.02	75.77 ± 4.66	85.28 ± 0.37	95.12 ± 1.33

Table 4b. v0.6. Descriptive test macro-F1, mean ± sample SD. All recipes use the same 1,600 test records per task; the 120 selected checkpoints are individual models, not an ensemble.

On checkerboard, Multi improves mean macro-F1 over Scalar by 5.53 percentage points and lowers mean BCE by 0.0564. On spiral, the corresponding changes are +7.11 points and −0.0754. Both BCE contrasts exclude zero in the adjusted seed and nominal adjusted record intervals. The macro-F1 changes are descriptive secondary outcomes. These results support the more flexible readout for these two tasks within the declared training budget; they do not isolate readout dimensionality from the accompanying additional parameters.

TrigR achieves the best mean F1 and lowest BCE on checkerboard and spiral: 89.57% and 92.05% F1. Its primary BCE improvements relative to both circuit recipes exceed the practical reference with adjusted intervals in the favorable direction. The stronger classical control therefore changes the interpretation of a circuit-only result. A circuit readout improvement is real in this study without being the best available classical prediction recipe.

For the noisy annulus and rare disks, rankings depend on the metric and several primary differences are small. Multi has the highest annulus observed-label F1, while TrigR has the lowest BCE. TrigG has the highest rare-disk F1, while Scalar has the lowest BCE. A single best F1 would omit these differences in confidence quality, uncertainty and task difficulty. Table 5 reports the complete primary family rather than selecting only favorable comparisons.

5.1 Primary contrasts and uncertainty

Task	First − second	Δ BCE	Adjusted seed CI	Adjusted record CI
Checker	Multi − Scalar	-0.0564	[-0.0920, -0.0209]	[-0.0833, -0.0286]
Checker	Scalar − TrigR	+0.2730	[+0.2388, +0.3072]	[+0.2401, +0.3041]
Checker	Multi − TrigR	+0.2166	[+0.2075, +0.2256]	[+0.1905, +0.2412]
Checker	TrigR − TrigG	-0.3915	[-0.3961, -0.3869]	[-0.4291, -0.3464]
Spiral	Multi − Scalar	-0.0754	[-0.1036, -0.0471]	[-0.1020, -0.0448]

Task	First − second	Δ BCE	Adjusted seed CI	Adjusted record CI
Spiral	Scalar − TrigR	+0.2524	[+0.2441, +0.2608]	[+0.2122, +0.2905]
Spiral	Multi − TrigR	+0.1770	[+0.1528, +0.2013]	[+0.1474, +0.2085]
Spiral	TrigR − TrigG	-0.4196	[-0.4224, -0.4169]	[-0.4603, -0.3675]
Annulus	Multi − Scalar	-0.0015	[-0.0105, +0.0074]	[-0.0118, +0.0090]
Annulus	Scalar − TrigR	+0.0075	[+0.0032, +0.0117]	[-0.0050, +0.0197]
Annulus	Multi − TrigR	+0.0059	[+0.0006, +0.0113]	[-0.0027, +0.0143]
Annulus	TrigR − TrigG	-0.0566	[-0.0590, -0.0541]	[-0.0894, -0.0255]
Disks	Multi − Scalar	+0.0011	[-0.0018, +0.0040]	[-0.0008, +0.0034]
Disks	Scalar − TrigR	-0.0014	[-0.0036, +0.0007]	[-0.0047, +0.0012]
Disks	Multi − TrigR	-0.0003	[-0.0025, +0.0019]	[-0.0026, +0.0016]
Disks	TrigR − TrigG	-0.0018	[-0.0022, -0.0013]	[-0.0046, +0.0016]

Table 5. v0.6. All 16 internally prespecified primary contrasts. Negative favors the first model. Nominal 99.6875% per-contrast intervals use the 16-contrast Bonferroni family. Seed intervals use five paired optimization seeds; record percentile intervals use 10,000 paired resamples conditional on the fitted checkpoints. These quantify different uncertainties and do not establish population replication.

The four checkerboard and four spiral contrasts agree in direction across both uncertainty views. For annulus and disks, some narrow seed intervals exclude zero while the corresponding record interval includes it. Small variation across training seeds on one split does not remove uncertainty about unseen records. Differences below the 0.01 BCE reference are not presented as practically established superiority.

5.2 Primary contrasts and secondary task scores

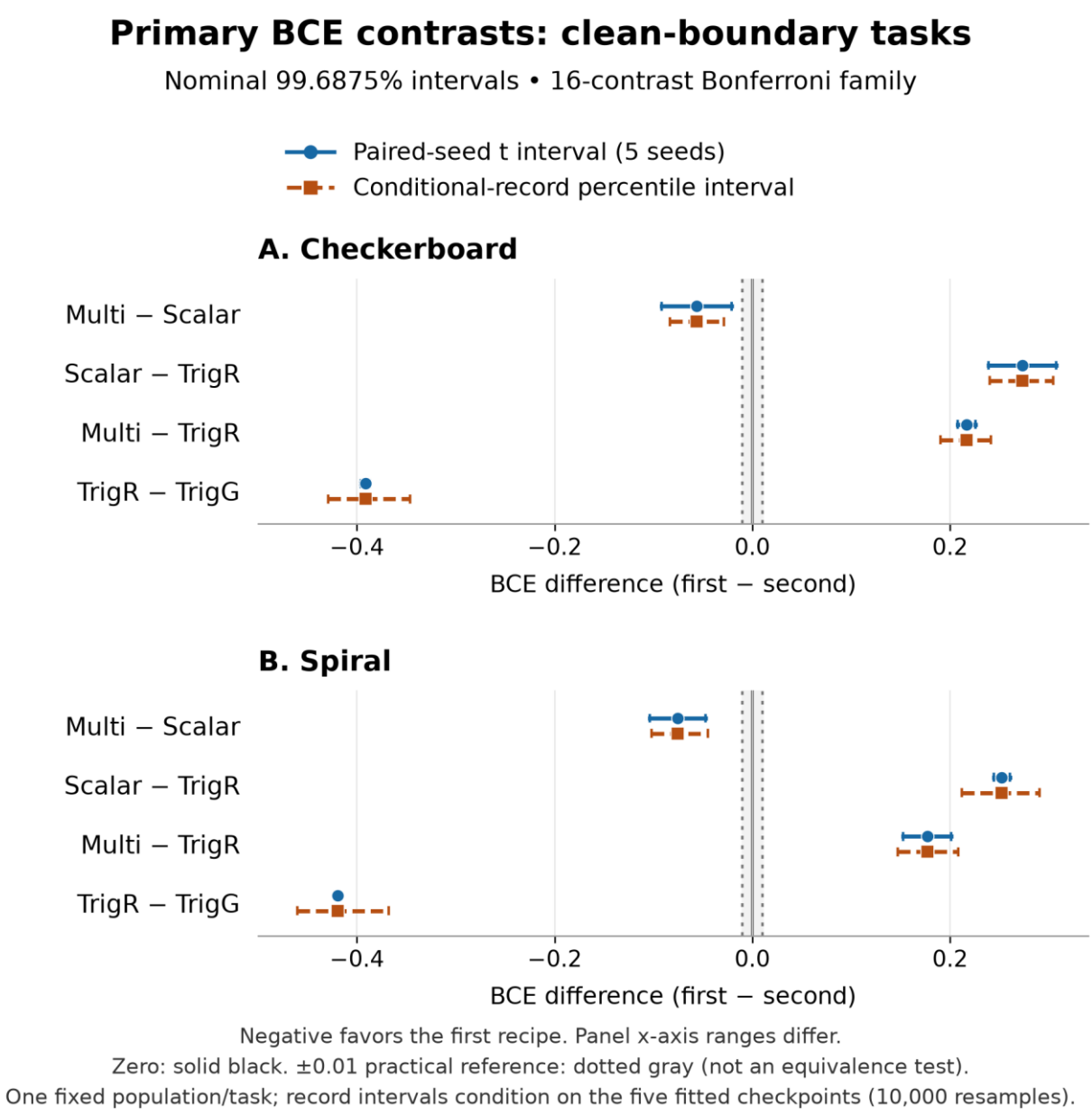


Figure 2a. v0.6. Primary BCE contrasts for the clean-boundary tasks, from full-precision retained records. Negative favors the first recipe. Seed t intervals and nominal conditional record-percentile intervals are shown separately, with task-specific horizontal scales. Zero and ±0.01 are displayed; five training seeds are not independent populations.

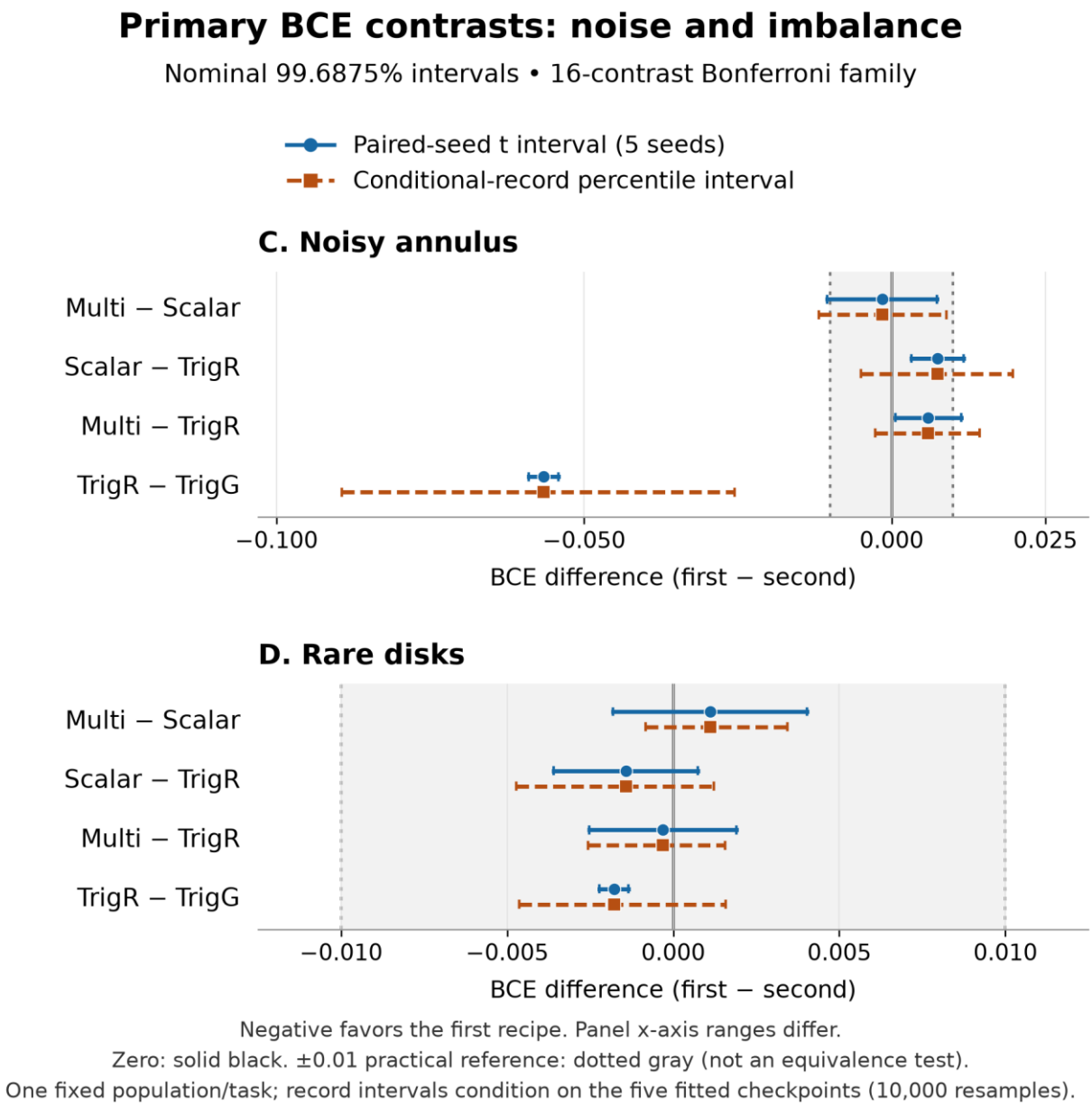


Figure 2b. v0.6. Primary BCE contrasts for noisy and imbalanced tasks. Both parts of Figure 2 retain the complete 16-contrast family. Scales differ by task; reference lines and interval types have the same meaning as in Figure 2a. Numerical values remain in Table 5.

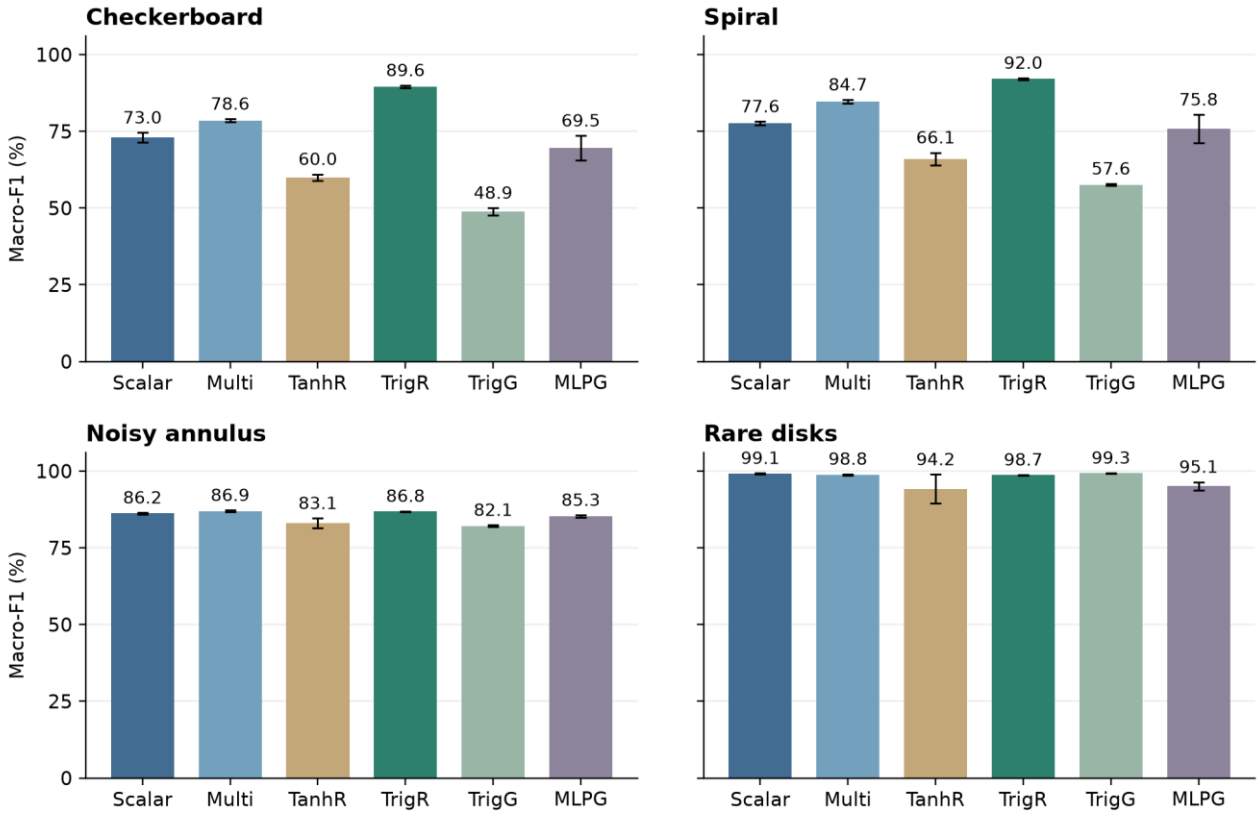


Figure 3. v0.6. Mean test macro-F1 with ±1 sample SD across five training seeds. All four panels share the 0–100% scale. Annulus uses observed noisy labels. Error bars describe training variability on a fixed test population and are not confidence intervals over new tasks.

The contrast between checkerboard and disks is informative. A model can score near 99% on a compact, smooth rare-disk target and remain much less successful on repeatedly alternating boundaries. The results therefore characterize particular model–task matches rather than a universal level of intelligence. The global Trig control's strong disk result also prevents interpreting every high routed score as evidence that routing was necessary.

The routed/global Trig difference is large on checkerboard and spiral, but routed Trig has 125 coefficients while the global model has 25. Both the piecewise partition and the larger total coefficient set can contribute. A later capacity-controlled study would need explicit equal-budget alternatives, multiple independently generated populations and separate router ablations. The current study provides the empirical reason for that work, rather than resolving it.

5.3 Noise, class balance and learning milestones

The annulus test split contains 176 flipped labels among 1,600 records (11%). A diagnostic with access to the known latent rule and true conditional probabilities 0.1/0.9 obtains 89.00% observed-label accuracy, 88.97% macro-F1 and 0.3471 BCE on this finite split. The population Bayes accuracy is 90% for independent 10% flips; it is not a finite-sample ceiling. This privileged reference is used only after release and is not a deployable trained competitor.

Model	Annulus latent F1 %	Disks positive recall %	Disks 97% reached	Median updates among reachers[1]
Scalar	95.98	98.40	5/5	750
Multi	96.85	97.97	5/5	600
TanhR	92.32	89.73	3/5	11250
TrigR	97.31	97.86	5/5	300
TrigG	90.94	98.61	5/5	900
MLPG	95.06	91.87	0/5	Not reached

Table 6. v0.6. Secondary diagnostics, not additional primary tests. [1]Median confirmation update among reachers only; 97% is validation macro-F1, confirmed at the third consecutive qualifying check. Nonreachers are censored at 15,000. Latent labels were excluded from learning and checkpoint selection.

Latent-label F1 shows that substantial annulus geometry is learned despite the noisier observed target. It does not replace the observed-label primary results. On rare disks, 187/1,600 test records are positive (11.69%). Positive recall exposes minority-class differences hidden by majority-heavy accuracy. Macro-F1 is useful here, but it is still accompanied by probability losses and class-specific diagnostics.

Only TrigR confirms 90% validation macro-F1 in all five runs on checkerboard and spiral; no model confirms 95% or 97% on either task. No model reaches 90% on noisy annulus. All full runs nevertheless continue to 15,000 updates. The rare-disk milestones quantify speed of attaining a validation target under a fixed schedule, not the number of steps needed to guarantee a future test score.

Native-logit BCE preserves the model ordering for each task. The largest relevant clipping difference is spiral TrigR: native mean 0.25158 versus common clipped mean 0.24828. Five wrong-clipped checkpoint–record occurrences contribute; these are not five independent new examples. The complete native-loss, calibration and class-quality summaries remain in the internal evidence.

5.4 Routing outcomes in v0.9

Table V9.1 retains every primary comparison. C_L has five favorable, three unfavorable and four inconclusive adjusted comparisons; three favorable intervals establish a gain beyond 0.01. Learned routing helps relative to fixed soft circuits on spiral and disks, but the G121 classical baseline is stronger than C_L on checkerboard, spiral and disks. C_L is stronger than G121 on annulus. A general superiority claim for learned circuit routing is therefore unsupported. The extra residual control and all other arms remain visible in Appendix E.

Table V9.1. All v0.9 primary common-BCE contrasts; negative favors C_L. Source [12], statistical analysis primary_contrasts.

Task	Contrast	Mean Δ	99.583333% interval	Outcome
checkerboard	C_L − C_S	-0.018135	[-0.041990, 0.005720]	Inconclusive
checkerboard	C_L − C_R	-0.003597	[-0.024373, 0.017179]	Inconclusive

Task	Contrast	Mean Δ	99.583333% interval	Outcome
checkerboard	C_L − G121	0.467345	[0.436036, 0.498654]	G121
spiral	C_L − C_S	-0.042174	[-0.057630, -0.026718]	C_L; gain >0.01
spiral	C_L − C_R	-0.024362	[-0.047379, -0.001345]	C_L
spiral	C_L − G121	0.252828	[0.226378, 0.279278]	G121
annulus	C_L − C_S	-0.003498	[-0.012289, 0.005293]	Inconclusive
annulus	C_L − C_R	0.001229	[-0.005458, 0.007915]	Inconclusive
annulus	C_L − G121	-0.022298	[-0.031476, -0.013120]	C_L; gain >0.01
disks	C_L − C_S	-0.022452	[-0.030175, -0.014730]	C_L; gain >0.01
disks	C_L − C_R	-0.009706	[-0.013059, -0.006353]	C_L
disks	C_L − G121	0.007245	[0.004393, 0.010097]	G121

Eight independent population means with two nested paired seeds, df7; 12-comparison Bonferroni family. Inconclusive does not mean equal. Practical gain requires the entire interval below−0.01.

The 12 secondary comparisons are shown separately in Appendix E at nominal 95% confidence. T_L improves on T_S for checkerboard, spiral and disks, with annulus inconclusive. Fixed soft C_S versus hard C_H improves annulus but worsens checkerboard and disks; spiral is inconclusive. These exploratory results add context without altering the primary multiplicity family. The descriptive Banknote means favor MLP4 (BCE 0.0165, macro-F1 0.9912) over G121 (0.3661,0.8240) and C_L (0.4078,0.7792), but MLP4 uses four inputs rather than two and differs in capacity. All 12 Banknote arms and seed ranges are retained in Appendix E.

5.5 Allocation outcomes in v0.9.5

None of the eight primary intervals in Table V95.1 lies wholly below−0.01. ETEL's mean BCE is slightly lower than fixed allocation on every task, but all four adjusted intervals cross zero. ETEL and halving have observed exact test-BCE ties on checkerboard, spiral and disks; annulus slightly favors halving with an interval crossing zero. All upper limits are below the registered+0.01 reference, which is not a formal equivalence claim or a guarantee on other candidate pools.

Table V95.1. All v0.9.5 primary native-BCE contrasts; negative favors ETEL. Source [13], PRIMARY_CONTRASTS.json.

Task	Contrast	Mean Δ	99.375% interval
checkerboard	ETEL − fixed	-0.004010	[-0.008685, 0.000664]
checkerboard	ETEL − halving	0.000000	[0.000000, 0.000000]
spiral	ETEL − fixed	-0.003136	[-0.007361, 0.001088]
spiral	ETEL − halving	0.000000	[0.000000, 0.000000]
annulus	ETEL − fixed	-0.001638	[-0.008052, 0.004775]
annulus	ETEL − halving	0.001597	[-0.000513, 0.003707]
disks	ETEL − fixed	-0.000733	[-0.001799, 0.000333]
disks	ETEL − halving	0.000000	[0.000000, 0.000000]

Eight independent population means with two nested paired seeds, df7; eight-comparison Bonferroni family. Zero-width observed ties describe these selected endpoints, not universal policy equality.

Table V95.2. Selected test performance and CPU consumption for every policy and task. Source [13], Tables 2,3,5.

Task / policy	Native BCE [95% CI]	Macro-F1 (0–1) [95% CI]	Native CPU s	Unused s
checkerboard / fixed	0.0531 [0.0499, 0.0564]	0.9821 [0.9806, 0.9837]	1,449.09	75.61
checkerboard / halving	0.0491 [0.0482, 0.0501]	0.9833 [0.9821, 0.9845]	713.82	810.88
checkerboard / ETEL	0.0491 [0.0482, 0.0501]	0.9833 [0.9821, 0.9845]	1,463.10	61.60
spiral / fixed	0.0820 [0.0744, 0.0897]	0.9664 [0.9622, 0.9706]	1,087.67	2.95
spiral / halving	0.0789 [0.0701, 0.0877]	0.9672 [0.9628, 0.9716]	705.96	384.65
spiral / ETEL	0.0789 [0.0701, 0.0877]	0.9672 [0.9628, 0.9716]	1,087.61	3.01
annulus / fixed	0.3740 [0.3643, 0.3837]	0.8828 [0.8780, 0.8876]	1,155.79	3.13
annulus / halving	0.3708 [0.3623, 0.3792]	0.8833 [0.8783, 0.8882]	1,008.91	150.00
annulus / ETEL	0.3724 [0.3643, 0.3804]	0.8816 [0.8768, 0.8863]	1,157.31	1.61
disks / fixed	0.0105 [0.0086, 0.0124]	0.9894 [0.9865, 0.9923]	1,196.77	2.68
disks / halving	0.0098 [0.0074, 0.0121]	0.9895 [0.9867, 0.9922]	737.90	461.55
disks / ETEL	0.0098 [0.0074, 0.0121]	0.9895 [0.9867, 0.9922]	1,197.08	2.37

Performance intervals are descriptive 95% population intervals (n = 8), not adjusted policy-effect intervals. See Appendix F for other secondary outcomes and cost uncertainties.

Halving uses less mean CPU on every task: 713.82, 705.96, 1008.91 and 737.90 seconds for checkerboard, spiral, annulus and disks, respectively, versus ETEL 1463.10, 1087.61, 1157.31 and 1197.08. Mean overruns are zero. Identical ceilings permit different consumption: capped halving leaves budget unused rather than reallocating it after the survivor reaches 200 rounds. These descriptive costs and the quality results do not demonstrate a training-efficiency advantage for ETEL. We retain halving as a strong benchmark without claiming universal superiority.

Full reference fits diagnose later validation improvement for one preselected seed per population. Misses for fixed/halving/ETEL are 4/0/0 on checkerboard, 7/0/0 on spiral, 5/5/4 on annulus and 8/0/0 on disks (each out of 8). They neither alter original selections nor establish hidden-test superiority; reference CPU lies outside the original policy budgets. G121 is selected throughout checkerboard and disks and throughout spiral at base or half rate. Annulus selects mixtures of C_S and C_L, with one fixed-policy G121 selection. All counts and secondary outcomes are retained in Appendix F.

6 Exact conversion and CPU deployment

Sections 6.1–6.3 describe conversion and CPU inference for the selected v0.6 predictors. Section 6.4 retains the separate v0.9 conversion evidence.

6.1 Finite function representation

Each coordinate appears in two RY(πxq) feature gates. Writing tq = exp(iπxq/2), each feature-gate entry is linear in tq and tq^{-1}. After the two uploads, amplitudes have powers −2, 0 or 2 in each tq; input-independent rotations and CNOTs introduce no new input frequencies. A Pauli expectation combines amplitudes and their conjugates, giving integer frequencies −2 through 2 in πxq. Any real affine combination of these expectations therefore lies in the 25-feature span of Equation 4 [5,6].

This construction specializes the classical Fourier-surrogate framework of Schreiber, Eisert and Meyer [6] to two-coordinate, two-upload affine logits. The contribution here is its controlled routed comparison, independent coefficient verification and measured deployment tradeoff.

The compiler evaluates each bank separately on the 5 × 5 nodes sj = −1 + 2j/5, j = 0,…,4. If Bja = ϕa(sj) and Z is the forced-bank logit grid, orthogonality gives the coefficients without regression or label use:

$$B^{\mathrm{T}}B \;=\; diag(5, 5/2, 5/2, 5/2, 5/2), \quad C \;=\; D\,B^{\mathrm{T}}\,Z\,B\,D, \qquad (6)$$
$$D \;=\; diag(1/5, 2/5, 2/5, 2/5, 2/5).$$

The five coefficient tables retain the original router and sigmoid. Thus conversion preserves the selected predictor's real-arithmetic logit per bank and its routed input-to-output function. It does not reconstruct a quantum state, create a new trained arm or expand the trained function class. The sigmoid probability itself need not have a finite Fourier expansion, and the hard-routed function need not be globally continuous.

6.2 Independent verification and numerical limits

Forty selected Scalar/Multi checkpoints are converted. Independent Laurent-polynomial gate propagation reconstructs coefficients without sampled input fitting or the projection solve. Validation and test records, off-grid fixtures, actual routing boundaries with $\pm 10^{-10}$ perturbations, and forced-bank checks test predictions. Across 159,560 circuit-conversion comparisons, the maximum probability discrepancy is 7.16×10^{-15}, maximum logit discrepancy 6.39×10^{-14} and the number of changed classifications is zero. The independent maximum coefficient difference is 2.13×10^{-14}.

$$E_{\mathrm{k}} \;=\; \Sigma_{\mathrm{a}=0}{}^{4}\,\Sigma_{\beta=0}{}^{4}\,|\Delta C_{\mathrm{a}\beta}{}^{(\mathrm{k})}| \;=\; \|vec(\Delta C^{(\mathrm{k})})\|_1, \;\; \Delta C^{(\mathrm{k})} \;=\; Cref^{(\mathrm{k})} \;-\; Cconv^{(\mathrm{k})}.$$
$$|zref(x) \;-\; zconv(x)| \;\le\; E_{\mathrm{r}(\mathrm{x})} \;\le\; max_{\mathrm{k}}\,E_{\mathrm{k}},$$
$$|\sigma(zref(x)) \;-\; \sigma(zconv(x))| \;\le\; E_{\mathrm{r}(\mathrm{x})}/4 \;\le\; (max_{\mathrm{k}}\,E_{\mathrm{k}})/4. \qquad (7)$$

For bank k, Cref and Cconv denote retained reference and converted coefficient tables, and E_k is the sum of absolute differences across all 25 entries. This is the vectorized entrywise 1-norm, not the induced matrix 1-norm. Every basis feature has magnitude at most one and the sigmoid derivative is at most one quarter. Equation 7 assumes the same router and tie handling. Applied to retained numerical tables, it excludes evaluation rounding and is not certification against an infinite-precision circuit oracle. The finite checks are not bitwise equality for all inputs. More uploads, changed input scaling or nonlinear heads can require another representation.

6.3 Classical inference performance

Representation	Batch 1	Batch 32	Batch 1,024
Scalar / compiled Scalar	5.00×	18.58×	5.80×
Multi / compiled Multi	5.32×	19.90×	6.15×

Table 7. v0.6. Median paired ratios of original statevector inference time to compiled inference time over 20 selected checkpoints per circuit type (four tasks × five seeds). Ratios exceed one when conversion is faster. They are hot CPU inference measurements, not training speedups or quantum-device comparisons.

The benchmark covers all 120 selected models, 40 compiled circuit representations and 40 equally optimized learned-Trig representations: 200 representations at three batch sizes. Each condition uses two warmups and 15 timed trials, for 1,200 warmups and 9,000 timed trials. Input fixtures are chosen by validation record ID without using labels. Method order rotates and reverses deterministically, and circuit/compiled order alternates. The benchmark runs after other study workers stop, in one process with one BLAS thread.

Representation	Batch-1,024 median µs/record	Numeric predictor bytes
Original Scalar → compiled	1.206 → 0.210	400 → 1,000
Original Multi → compiled	1.274 → 0.208	560 → 1,000
Optimized learned TrigR	0.208	1,000
TanhR	0.219	1,320
MLPG	0.0526	1,032

Table 8. v0.6. Pooled checkpoint-median timing summaries and float64 predictor payloads. Add 80 numerical centroid bytes for routed methods; serialization and process memory are not included. Pooled medians are descriptive and are not used to reconstruct Table 7's paired ratios.

Conversion takes approximately 1 ms per selected checkpoint in the recorded workload and increases coefficient storage relative to the compact circuit parameters. It trades a larger explicit classical representation for less repeated statevector work. Learned Trig coefficients use the same optimized evaluation kernel, so their similar latency demonstrates that the compiled speed is available to a directly trained classical control as well. The original small global MLP is faster still in the reported large-batch summary.

Raw input routing, feature construction and sigmoid evaluation are inside the timed call; loading and conversion are outside it and separately recorded. No persistent design-matrix cache hides feature work. Later annulus/disks blocks were approximately 1.7–1.8× slower across implementations, indicating session drift. Task blocks are sequential, so that difference cannot be attributed causally to task complexity. These measurements support a local representation tradeoff, not a universal hardware-independent speed claim.

6.4 Separate conversion evidence in v0.9

The v0.9 conversion study checks 276 circuit models comprising 1,380 bank cores, each represented by 25 trigonometric coefficients [12]. It preserves preprocessing, gates, residuals and the mixture of

expert probabilities; the whole soft-routed predictor is not one global 25-term logit. Across 2,484 matched endpoint/batch/session comparisons, the converted implementation was faster in all retained cells. Three sessions include 123,120 timed trials and 16,416 warm-up counters; individual warm-up records were not retained. These are technical repeated measurements on this implementation, not independent predictive replicates or a hardware-independent speed guarantee. They do not measure v0.9.5 training efficiency.

Independent scoring used a separately implemented forward oracle and statistics, while a reviewed recovery shared SciPy expit for a post-fit argmax diagnostic; tolerances were not changed and all 636 endpoints passed. An independent conversion metadata failure occurred before numerical loops; its reviewed repair checked all 276 models. Such integrity results establish the stated implementation checks, not new predictive outcomes. The original v0.6 conversion and simulator measurements above remain separate.

7 Local noise and measurement diagnostics

This section reports the v0.6 local simulator and measurement diagnostics. Shot replicates and shared fixture inputs are technical units within that study, not additional independent populations.

The readiness capsule remains entirely classical. It uses 40 selected circuit checkpoints and the 32 lowest-ID validation records per task. Scalar uses one joint XX measurement setting to estimate X_o. Its implemented setting rotates both qubits for readout, and both rotations are charged. Multi uses joint XX and ZZ settings to obtain X_o, X_1, Z_o, Z_1 and Z_oZ_1. Moments within a setting are derived from the same joint outcome counts; they are not sampled independently. Increasing readout flexibility therefore changes a prospective measurement budget.

Condition	What is represented	Interpretation
Ideal	Classical statevector/density evolution and exact joint probabilities	Reference implementation agreement
Illustrative logical noise	1-qubit depolarization 0.001; CX depolarization 0.01; readout flip 0.01	Declared sensitivity scenario, not device calibration
Archived Manila snapshot	Local fake-backend snapshot dated 27 May 2024	Historical noise/compilation fixture; not current calibration
Finite shots	1,024 or 8,192 shots per setting; five technical replicates	Conditional sampling variability of fixed fitted models

Table 9. v0.6. CPU simulator conditions. A historical device snapshot supports offline tests but does not make them physical quantum experiments [9].

Compilation retains five backend wires: logical q_o maps to wire/physical qubit 1 and logical q_1 to wire/physical qubit 0. Optimization level is 1, transpiler seed 690017, with no routing swaps; counts are reordered to logical order. The 32 lowest-ID validation fixtures per task produce 1,920 setting circuits, 5,760 density evaluations, 1,800 joint-count arrays and 1,200 predictor arrays; multinomial draws represent 265,420,800 outcomes. The actual Aer smoke uses the first 8 of those 32 IDs for all 40 checkpoints: 480 distinct setting circuits, each executed under ideal and archived-Manila noise.

Thus 960 simulator experiments produce 983,040 simulated shots at 1,024 per setting. Physical shots are zero.

Independent density/count replay agrees to maximum probability difference 3.56×10^{-14} and reproduces retained counts and bias–variance calculations. It does not constitute an independent repeat of transpilation or all Aer execution. The small validation-fixture capsule describes probability distortion and threshold sensitivity, not a fresh held-out hardware F1 study. Technical shot replicates are not additional independently trained models.

Noise can introduce systematic probability changes that additional shots alone cannot remove. More shots reduce the variance of linear moment estimates. After the nonlinear sigmoid, finite-sample probability variance need not decrease monotonically; the recorded ideal variances were lower at 8,192 shots in each displayed case. The affine head can amplify moment errors. Neither the illustrative noise rates nor the archived Manila snapshot predicts a future device's behavior; any future physical comparison requires contemporaneous hardware provenance and a separately registered measurement protocol.

7.1 Measured local sensitivity

Task	Model	Logical noise MAE (disagreements)	Archived noise MAE (disagreements)
Checker	Scalar	0.0355 (14)	0.0493 (15)
Checker	Multi	0.0336 (8)	0.0466 (11)
Spiral	Scalar	0.0109 (0)	0.0152 (1)
Spiral	Multi	0.0167 (3)	0.0213 (4)
Annulus	Scalar	0.0135 (0)	0.0193 (0)
Annulus	Multi	0.0137 (1)	0.0225 (2)
Disks	Scalar	0.0026 (1)	0.0029 (1)
Disks	Multi	0.0032 (1)	0.0053 (1)

Table 10. v0.6. Mean absolute probability distortion from the ideal predictor; parentheses count prediction disagreements among 160 checkpoint–fixture pairs per cell (32 shared inputs × five checkpoints). These are conditional diagnostics, not F1 or 160 independent records. Ideal density parity produced no prediction disagreements.

Task	Model	1,024/setting: variance	Total shots/prediction	8,192/setting: variance	Total shots/prediction
Checker	Scalar	0.008411	1,024	0.000822	8,192
Checker	Multi	0.005303	2,048	0.000996	16,384
Spiral	Scalar	0.016039	1,024	0.002601	8,192
Spiral	Multi	0.012678	2,048	0.001416	16,384

Task	Model	1,024/setting: variance	Total shots/prediction	8,192/setting: variance	Total shots/prediction
Annulus	Scalar	0.000516	1,024	0.000075	8,192
Annulus	Multi	0.000594	2,048	0.000061	16,384
Disks	Scalar	0.000898	1,024	0.000210	8,192
Disks	Multi	0.001000	2,048	0.000099	16,384

Table 11. v0.6 empirical probability variance over five technical shot replicates per checkpoint–fixture pair, averaged over 160 pairs. Scalar uses one setting and Multi two: equal per-setting budgets are not equal total budgets. The v0.6 report proposed a matched-total-shot follow-up. This table retains its original diagnostics, whose total shot budgets were not matched across readout settings.

These retained v0.6 diagnostics do not establish the outcome of a later matched-total-shot study. No new simulator experiment was run for this synthesis.

The recorded ideal variance is lower at the larger shot budget in each displayed case. This does not remove the noise-induced distribution shift in Table 10. Circuit readout quality, head sensitivity, total measurements and task-score uncertainty must all be assessed together in a future physical comparison.

8 Discussion, limitations and conclusion

8.1 What the evidence supports

The earlier v0.6 evidence supports richer readouts for the declared shallow circuit on checkerboard and spiral, while its freely trained routed trigonometric control remains stronger. v0.9 adds population-level evidence that learned routing is useful only conditionally: C_L has five favorable and three unfavorable primary comparisons, with four inconclusive. G121's strong results on three tasks and the Banknote controls prevent a blanket claim for the circuit parameterization. These studies do not separate expressivity limits from optimization difficulty or establish each family's best attainable tuning.

The allocation experiment provides a distinct result. The registered ETEL policy preserved recoverable execution but did not establish a practical native-BCE or training-efficiency advantage over fixed or halving. Exact ties on three tasks coexist with markedly different CPU consumption. Successful checkpoint/replay machinery is an engineering property, not evidence that its allocation utility is optimal. Classical conversion remains a deployment option for selected functions; its inference timing cannot be used as evidence of ETEL training savings.

8.2 What remains unresolved

v0.6 seed intervals remain conditional on one population and router per task; n5 limits assessment of t assumptions. v0.9 and v0.9.5 improve population replication to n = 8 per task with two paired seeds, but remain small synthetic binary studies. Their eight population means, not 16 seeds or many checkpoints, are the inferential units. The Banknote check adds one real-data split, not a replicated realistic benchmark. Out-of-distribution, higher-dimensional and changing-distribution generalization remains unestablished.

Within each study, recipes differ in parameter count, activation, function class and optimization geometry. In v0.6, routed versus global Trig changes allocated coefficients; the checkerboard frequency-three target also exceeds the original global logit basis. In v0.9, learned gates and residuals add parameters and soft routing evaluates all banks. In v0.9.5, the candidate pool and utility rule are fixed, halving's single-survivor cap can leave CPU unused, and the development cost amendment is retained. The result concerns that registered experiment, not every possible allocator, circuit or classical model. The 60-round first review was prespecified; this study did not compare alternative review fractions or establish 30% as optimal.

Adaptive validation reuse is a limitation of v0.9.5: the same validation set guides repeated allocation decisions and final checkpoint selection, and replacement development reuses the same development populations. The independent hidden tests evaluate the resulting frozen procedure, but do not establish generalization beyond these task families or remove development overfitting. A bounded late-improver probe cannot guarantee that every useful candidate is retained.

The full architecture remains incompletely tested. v0.9 learned input routing and v0.9.5 allocation over whole candidates do not validate integrated semantic priority, inter-stage representation updates, continual adaptation or retention. Earlier text-classification evidence is limited and provides no language-generation claim. Appendix D retains these gaps. High scores, exact conversion and audit PASS do not jointly prove architectural validity.

CPU timings depend on implementation, batching and host activity; overlapping process walls cannot be summed into project elapsed time. v0.9.5 retains failures and conservative crash charges, while earlier auxiliary lifetimes and isolated controller/serialization costs are incomplete. No exact all-process cost total is manufactured. Peak memory, energy and physical-QPU costs are not established by these results. Simulator noise and shot tests remain local preparation; device calibration, queue delay, measured output quality and end-to-end costs require actual hardware data.

8.3 Further classical studies

Further classical work would use the accepted allocation code with fresh datasets, passive metadata qualification and a calibrated allocator compared with halving and bridge controls. A separate proposed pilot concerns bounded CPU inference and retrieval with a frozen language model, synthetic fixtures and bounded text evaluation. Both remain proposals within the future research programme, without reported results. Feasibility, resource requirements and compact retention need to be measured. Neither proposal establishes text training or physical-QPU performance.

8.4 Planned physical comparison in the v2.0 research programme

The planned private v2.0 research programme includes a separately specified classical-versus-quantum comparison. Its first physical study would evaluate hardware inference using classically trained frozen checkpoints, matched inputs, retained routing and classical postprocessing. It would compare ideal simulation, compiled prediction, classical controls and measured device output. Quantum hardware training remains a separate future question. Internal correctness checks, reproducible runs and safety review must precede any release of resulting findings.

A separate protocol must specify checkpoint/input selection, backend and calibration provenance, layouts, measurement grouping, total shots, mitigation, retries and costs before outcomes. Any hardware task score needs an adequate held-out evaluation set. Classical training, queueing, device execution and postprocessing costs must be reported separately. No physical-device results are present in this manuscript.

8.5 Conclusion

The three studies support a restrained account of Q-Pathformer's classical components. v0.6 improves a shallow circuit readout but retains stronger classical controls on difficult tasks and a verified deployment conversion. v0.9 shows task-dependent routing benefits with favorable, unfavorable and inconclusive comparisons. v0.9.5 finds no demonstrated practical ETEL gain and identifies capped halving as a strong quality-and-cost benchmark for the fixed candidate pool. These outcomes motivate precise classical tests and a separate physical comparison; they establish neither integrated general capability nor quantum advantage.

8.6 Availability and preparation statement

This v1.0 collection remains a private draft for review. It contains this paper, the architecture companion, original architecture and ten study reports. Release documents specify availability and individual licences; participation policies are in the separate Research Approach. AI assistance supported drafting, evidence extraction, code and document preparation. Internal checks are not external replication or peer review.

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Project records and the collection papers are identified in Appendix C and the release index. Repository links require access while the repository is private. The literature review is focused on the evaluated mechanisms and does not establish exhaustive priority or novelty. Each external source has its own bibliographic entry; the original architecture and all ten supporting studies are separately identified.

Appendix A Early development history from v0.1 to v0.6

The research developed through six internal study versions, v0.1–v0.6. Each contains multiple model, seed and task comparisons. “Six versions” does not mean six individual tests, six model baselines or six independent replications. Different tasks, controls and objectives were used across the series; routing was held fixed in some later versions while readouts or predictors changed.

The collection includes v0.1 [16], v0.2 [17], v0.3 [18], v0.4 [19], v0.5 [20], v0.6 [2], v0.7 [21], v0.8 [22], v0.9 [12] and v0.9.5 [13]. Each study remains separate evidence, without cross-version pooling.

Version	Experimental focus	Main lesson and limitation
v0.1	Initial text and circuit baselines: shared/content text adapters; assigned/shuffled depths and fixed-depth controls.	Established component baselines and classical controls; semantic priority and the integrated architecture were not tested.
v0.2	Text-route stability and parameter sharing: current/frozen content and hash text routing; depth-specific circuit banks.	Untied banks improved the original circuit schedule; hashing did not resolve a text-quality gain. Capacity and exposure remained intertwined.
v0.3	Shallow circuits and routing: input-hash and k-means banks, assigned-depth, global and ensemble controls.	Geometric routing helped on the circle task; classical partitioning contributes structure, and the larger MLP remained stronger overall.
v0.4	Routing versus circuit contribution: global, hash and k-means modes across circuit, MLP and Trig families.	Matched-router comparisons exposed strong classical alternatives; useful routing did not establish a uniquely beneficial circuit contribution.
v0.5	Flexible readouts and capacity: k-means held fixed for routed recipes, with global controls; no hash arm.	Readouts improved the declared recipes, but the legacy comparison also changed training loss. Strong classical controls remained important.
v0.6	Harder tasks and CPU deployment: fixed k-means circuit/classical models versus global controls; no new hash arm.	Readout gains were task dependent; exact conversion sped inference. Routed Trig remained stronger on the difficult clean boundaries.

Table A1. Retrospective development history. The rows summarize the evidence and its limits; they are not a pooled performance ranking. Results from AG News, DBpedia and synthetic geometry are not directly comparable. Source identities and hashes are retained in REVISION_HISTORY_EVIDENCE.json.

The series identified useful spatial partitioning, task-dependent benefits from flexible readouts and an efficient classical representation of selected circuit predictors. It also exposed unresolved text-routing gains, strong conventional controls, task-dependent rankings, and capacity or optimization confounding. These observations informed later designs without demonstrating uniform progress or complete architectural validation.

The v0.6 numerical evidence retains its own task definitions, controls and uncertainty. Later v0.9 and v0.9.5 findings are presented separately in Sections 4.5–4.6 and 5.4–5.5 and Appendices E/F. Historical runs are not added to any later sample size. Appendix D lists broader capabilities that remain untested.

Appendix B Secondary probability diagnostics

Task	Model	Accuracy %	Brier	ECE10
Checker	Scalar	73.07	0.1897	0.0811
Checker	Multi	78.59	0.1656	0.0964
Checker	TanhR	60.21	0.2291	0.0353
Checker	TrigR	89.57	0.0877	0.0700
Checker	TrigG	49.24	0.2517	0.0355
Checker	MLPG	69.56	0.1928	0.0488
Spiral	Scalar	77.60	0.1577	0.0624
Spiral	Multi	84.71	0.1262	0.0874
Spiral	TanhR	66.06	0.1947	0.0603
Spiral	TrigR	92.05	0.0658	0.0613
Spiral	TrigG	57.62	0.2386	0.0753
Spiral	MLPG	75.77	0.1529	0.0646
Annulus	Scalar	86.24	0.1209	0.0732
Annulus	Multi	86.96	0.1202	0.0771
Annulus	TanhR	83.12	0.1331	0.0458
Annulus	TrigR	86.88	0.1175	0.0721
Annulus	TrigG	82.15	0.1389	0.0536
Annulus	MLPG	85.29	0.1253	0.0758
Disks	Scalar	99.64	0.0036	0.0071
Disks	Multi	99.53	0.0039	0.0074
Disks	TanhR	97.56	0.0163	0.0150
Disks	TrigR	99.45	0.0042	0.0063
Disks	TrigG	99.71	0.0042	0.0106
Disks	MLPG	97.97	0.0154	0.0141

Table B1. Means across five selected checkpoints; descriptive secondary outcomes. Brier uses raw probabilities. ECE uses ten fixed bins of predicted positive-class probability; it is sensitive to binning and finite-sample variation. No post-test calibration is fitted. Source: sealed confirmation statistics and manuscript results_evidence.json.

Appendix C Reproducibility and evidence map

The retained v0.6 delivery manifest records 20,859 payloads and 2,229,408,138 bytes. This editorial preparation does not repeat its historical seal checks. For the follow-up studies, the findings derive from accepted report and statistical artifacts. v0.9.5 main and finite-supplement verification preceded final report delivery. Integrity checks and prescribed-action replays have their stated scope and are not independent repeat training.

The release index maps each study report to its own Study, Documentation and Code assets. The relative evidence paths below describe the retained study layout; the full private evidence tree is not asserted to be included in the consolidated paper download. Owner records preserve historical workstation locations, immutable identifiers and relocation caveats. Historical verification does not certify a later expanded inventory or a separate archival copy.

Evidence	Location relative to the v0.6 root
Design and statistical rules	governance/STUDY_PLAN.json; ANALYSIS_PROTOCOL.json; DATA_REGISTRATION.json
Deployment and simulator rules	governance/DEPLOYMENT_PROTOCOL.json; LOCAL_READINESS_PROTOCOL.json
Hardware and dependencies	governance/RUNTIME_MACHINE.json; RUNTIME.json; observed and locked requirements
Models, routing and training	quantum/models.py; circuits.py; routers.py; study.py; prepare_data.py
Official confirmation results	quantum/confirmation_evaluation/summary.json and per-population results
Independent numerical replay	analysis/independent_replay/confirmation_replay/summary.json
Conversion and timing	quantum/deployment_confirmation/summary.json and raw artifacts
Noise and finite shots	quantum/local_readiness_confirmation/finite_shot_summary.json
Independent readiness / conversion checks	analysis/classical_review/confirmation_readiness_audit and confirmation_deployment_audit
Learning and resource accounting	analysis/learning_resources/stage_B_v01/summary.json and endpoint_metrics.csv

The preserved manuscript sources and supporting review records retain full-precision tables, figures, reconstruction records and document change history. All original numerical tables, four scientific figures and native equations are retained in this edition. Reproduction requires a verified writable copy, compatible retained dependencies and fresh output folders; immutable originals remain unchanged.

v0.9.5 reproduction guide

Official scoring, independent numerical review, the report, package verification and delivery closeout are recorded as complete in the retained study records. The frozen reproducibility/README.txt

describes an earlier handoff; its pending steps and placeholder approvals are historical. The retained reproduction index and report errata identify final accepted records and exact historical scoring commands. Those commands document provenance and do not authorize rerunning the archived ledger. Public availability is limited to the materials identified in each approved release.

Table C2 maps the principal v0.9.5 records relative to S, the evidence/ subdirectory of the study delivery. Exact registered populations and seed streams, the retained exclusion inventory and split rules govern regeneration; choosing new seeds creates a new experiment. The companion index supplies immutable identifiers and identifies the original runtime and current path mappings.

Table C2. v0.9.5 data code and accepted result records

Evidence	Locations relative to S
Registered population and training seeds	S\governance\RUN_MATRIX.json
Generator, namespace derivation and data rules	S\data\prepare_data.py; S\data\register_protocols.py; S\governance\DATA_PROTOCOL.json
Sealed data and training-only inputs	S\data\generated\DATA_SEAL.json; S\data\generated\TRAINING_INPUT_MANIFEST.json
Accepted matrix and migration	S\governance\development_cost_revision01\RUN_MATRIX_COSTREV01.json; MATRIX_MIGRATION_01.json in the same directory
Cost amendment and descriptive targets	S\governance\ROOT_COST_AMENDMENT_ACCEPTANCE_01.json; S\governance\MILESTONE_TARGETS_COSTREV01_v02.json
Author and independent scorer versions	S\analysis\author_v095\evaluate_v095.py; S\analysis\independent_scoring_v095_recovery01\replay.py
Scoring scientific source/protocol freezes	S\governance\ROOT_AUTHOR_EVALUATION_RELEASE_01.json; S\governance\ROOT_INDEPENDENT_SCORING_RELEASE_01.json
Historical scoring launch approvals	S\governance\ROOT_AUTHOR_SCORING_LAUNCH_01.json; S\governance\ROOT_INDEPENDENT_SCORING_LAUNCH_01.json
Committed results and independent scoring replay	S\analysis\author_scoring_v01\AUTHOR_EVALUATION_COMMIT.json; S\analysis\independent_scoring_v01\INDEPENDENT_SCORING_REPLAY.json
Four clean numerical replays and copy provenance	S\governance\ROOT_CLEAN4_JOINT_ACCEPTANCE_01.json; S\governance\ROOT_CLEAN4_PHYSICAL_COPY_ACCEPTANCE_01.json
Runtime, recovery rules and historical guide	S\governance\NUMERICAL_ENVIRONMENT_01.json; S\governance\REPRODUCTION_PROTOCOL.json; S\reproducibility\README.txt
Report build and native-render evidence	S\report_work\final_report\build_v02\BUILD_RECEIPT.json; ROOT_REPORT_OPERATION_COMPLETION_01.json in the same directory

The preserved runtime is stored under runtime/venv and runtime/base_python at the study root. Copies, exact-input replay and tolerance-based scoring checks do not establish a fresh installation or

a fully autonomous repeat on a new machine. New output paths and any absolute-path migration require a separate reproduction procedure; neither the historical approvals nor a file-integrity seal supplies that procedure.

v0.9.5 managed workers comprise 475 jobs and 487 retained attempts, with 316,492.40625 seconds observed native CPU and 316,507.55198 charged CPU. The 15.14573-second excess is conservative charging on six engineered crashes. These totals exclude coordinators, audits, scoring and reporting and are not a complete project total. Appendix F retains stage costs and failures. Packaging and closeout remain separate administrative costs, without being added to scientific compute.

Administrative token counters, account balances and delivery accounting are retained in the private owner records. They are not scientific compute measurements or project-attributed billing and are excluded from the experimental resource totals.

The study accounting distinguishes verified-main work, supplement verification, report delivery and later manuscript preparation. These administrative stages remain separate from the managed-worker CPU measurements reported above and in Appendix F.

The supplied methods and evidence support internal inspection and the specific numerical replay checks described here. Independent external reproduction, fresh-install portability and hardware-independent timing have not been established. Each replay retains its stated scope.

Appendix D Architectural scope and untested components

The architecture proposal [1] is broader than the evaluated mechanism. Its Q-Sphere, Cube and Sphere stages describe functional roles in prioritization, representation and routing, and adaptation. These stages are not three physical quantum states. The present results establish behavior for specified classical predictors and numerical two-qubit simulations; they do not establish end-to-end coordination of the proposed stages.

Priority and state coordination

v0.9 tests learned input gates and v0.9.5 tests a registered suitability allocator, providing bounded evidence rather than validation of semantic priorities. ETEL did not demonstrate practical test-BCE or training-efficiency gains over the registered comparators. Earlier metadata-assigned depths and these allocation rules do not establish real semantic class ratings or end-to-end coupling of priority, representation, routing and adaptation. Those mechanisms need separately specified objectives and controls.

Sequential adaptation and safeguards

The proposed sequence of representation updates, priority or class updates, and task-specific adaptation has not been comprehensively tested. Replay, retention constraints and protection against catastrophic forgetting require sequential tasks and explicit before-and-after measurements. Proposed acceptance, rejection and rollback policies need tests of their triggers, recovery behavior and effects on retained performance. Good performance on an independent static classification dataset does not establish those capabilities.

Scale and additional workloads

Earlier text-classification studies provide evidence for a limited supervised setting; they do not establish language-generation quality or full-scale LLM training. Systematic evaluation of higher-dimensional inputs, larger models and more banks remains open. Realistic changing distributions and multimodal integration also need evaluation. Their accuracy, routing overhead, memory use and end-to-end training cost must be measured directly. Parameter counts and small-task F1 alone cannot establish scaling or deployment suitability.

Physical quantum execution

Statevector, density-matrix and finite-shot results in v1.0 were computed classically. They do not establish physical-device accuracy, calibration robustness, end-to-end quantum execution cost or quantum advantage. A physical comparison within the planned private v2.0 programme would require measured hardware results and reproducibility code under a separately specified protocol before any publication. Exact conversion of the present shallow logits does not establish efficient classical simulation of arbitrary larger quantum systems.

The eight trainable circuit angles in each bank are not eight semantic priority classes. The tested five-bank and six-candidate mechanisms do not validate a semantic class policy. The proposed follow-up experiments remain future classical studies, and the physical-QPU comparison remains separate future work. No new experiment is executed or published by this manuscript revision.

Appendix E Full v0.9 follow-up results

All synthetic arms and the Banknote check are retained here. Synthetic SDs are sample SDs of eight population means after averaging paired seeds, not uncertainty across checkpoints. Banknote ranges span five optimization seeds on one split and are not population confidence intervals. Sources: [12], accepted reviewed report Tables labelled by task and STATISTICAL_ANALYSIS.json.

Table E1. v0.9. Checkerboard nine arms.

Arm	BCE	BCE SD	Macro-F1 (0–1)	F1 SD	Accuracy
C_H	0.5012	0.0260	0.7801	0.0268	0.7802
C_L	0.5156	0.0224	0.7569	0.0313	0.7575
C_R	0.5192	0.0133	0.7532	0.0208	0.7536
C_S	0.5337	0.0137	0.7408	0.0185	0.7414
G121	0.0482	0.0046	0.9833	0.0038	0.9833
T_H	0.2743	0.0152	0.9045	0.0091	0.9045
T_L	0.3091	0.0136	0.8971	0.0105	0.8971
T_R	0.3626	0.0149	0.8832	0.0107	0.8832
T_S	0.3747	0.0164	0.8795	0.0128	0.8795

Table E2. v0.9. Spiral nine arms.

Arm	BCE	BCE SD	Macro-F1 (0–1)	F1 SD	Accuracy
C_H	0.3648	0.0713	0.8629	0.0359	0.8630
C_L	0.3339	0.0189	0.8598	0.0157	0.8600
C_R	0.3582	0.0112	0.8453	0.0145	0.8458
C_S	0.3760	0.0128	0.8339	0.0199	0.8342
G121	0.0810	0.0104	0.9650	0.0036	0.9650
T_H	0.2357	0.0437	0.9229	0.0180	0.9229
T_L	0.2175	0.0091	0.9185	0.0109	0.9185
T_R	0.2361	0.0120	0.9219	0.0111	0.9220
T_S	0.2521	0.0164	0.9181	0.0128	0.9182

Table E3. v0.9. Annulus nine arms.

Arm	BCE	BCE SD	Macro-F1 (0–1)	F1 SD	Accuracy
C_H	0.3799	0.0193	0.8805	0.0111	0.8808
C_L	0.3564	0.0224	0.8902	0.0094	0.8904
C_R	0.3552	0.0228	0.8899	0.0095	0.8901
C_S	0.3599	0.0245	0.8901	0.0099	0.8903

Arm	BCE	BCE SD	Macro-F1 (0–1)	F1 SD	Accuracy
G121	0.3787	0.0225	0.8779	0.0103	0.8782
T_H	0.3760	0.0217	0.8843	0.0085	0.8846
T_L	0.3488	0.0232	0.8954	0.0096	0.8956
T_R	0.3472	0.0251	0.8958	0.0091	0.8960
T_S	0.3482	0.0238	0.8960	0.0088	0.8962

Table E4. v0.9. Disks nine arms.

Arm	BCE	BCE SD	Macro-F1 (0–1)	F1 SD	Accuracy
C_H	0.0168	0.0023	0.9856	0.0041	0.9943
C_L	0.0157	0.0021	0.9889	0.0048	0.9956
C_R	0.0254	0.0020	0.9874	0.0033	0.9950
C_S	0.0382	0.0039	0.9788	0.0087	0.9919
G121	0.0085	0.0027	0.9924	0.0030	0.9970
T_H	0.0141	0.0029	0.9887	0.0049	0.9955
T_L	0.0133	0.0013	0.9915	0.0039	0.9966
T_R	0.0149	0.0014	0.9925	0.0034	0.9970
T_S	0.0158	0.0012	0.9923	0.0034	0.9970

Table E5. v0.9. Banknote twelve arms.

Arm	BCE	BCE range	Macro-F1 (0–1)	F1 range	Accuracy
C_H	0.4183	0.3953 to 0.4361	0.7742	0.7597 to 0.7959	0.7774
C_L	0.4078	0.3895 to 0.4320	0.7792	0.7569 to 0.7907	0.7825
C_R	0.4217	0.4110 to 0.4331	0.7499	0.7387 to 0.7632	0.7526
C_S	0.4238	0.4136 to 0.4409	0.7578	0.7499 to 0.7885	0.7599
G121	0.3661	0.3539 to 0.3810	0.8240	0.7864 to 0.8408	0.8263
LR2	0.4423	0.4392 to 0.4436	0.7496	0.7464 to 0.7544	0.7547
MLP2	0.4142	0.3958 to 0.4241	0.7558	0.7358 to 0.7780	0.7577
MLP4	0.0165	0.0019 to 0.0279	0.9912	0.9889 to 1.0000	0.9912
T_H	0.4028	0.3972 to 0.4128	0.7605	0.7523 to 0.7706	0.7635

Arm	BCE	BCE range	Macro-F1 (0–1)	F1 range	Accuracy
T_L	0.3945	0.3905 to 0.3989	0.7769	0.7706 to 0.7811	0.7803
T_R	0.4146	0.4112 to 0.4169	0.7787	0.7745 to 0.7815	0.7818
T_S	0.4176	0.4165 to 0.4192	0.7733	0.7643 to 0.7780	0.7759

Table E6. All 12 exploratory v0.9 secondary common-BCE contrasts.

Task	Contrast	Mean Δ	Nominal95% interval
annulus	C_L − C_H	-0.023465	[-0.032153, -0.014778]
annulus	T_L − T_S	0.000625	[-0.001622, 0.002872]
annulus	C_S − C_H	-0.019967	[-0.029613, -0.010322]
checkerboard	C_L − C_H	0.014407	[-0.005973, 0.034787]
checkerboard	T_L − T_S	-0.065570	[-0.077288, -0.053853]
checkerboard	C_S − C_H	0.032542	[0.016743, 0.048341]
disks	C_L − C_H	-0.001130	[-0.002041, -0.000220]
disks	T_L − T_S	-0.002516	[-0.002885, -0.002147]
disks	C_S − C_H	0.021322	[0.016909, 0.025735]
spiral	C_L − C_H	-0.030902	[-0.087122, 0.025318]
spiral	T_L − T_S	-0.034584	[-0.047501, -0.021667]
spiral	C_S − C_H	0.011272	[-0.046788, 0.069331]

Negative favors the first arm. These unadjusted secondary intervals cannot be promoted to corrected primary claims. Different families, populations and BCE conventions prevent pooling with v0.6 or v0.9.5.

Appendix F Full v0.9.5 outcomes and resource limits

The following tables preserve all selected-model secondary metrics, their descriptive 95% population intervals, policy costs and validation-reference diagnostics. Each policy/task uses eight independent population means with paired seeds. All displayed conditional metrics use eight complete population means; no post-test tuning, calibration or new inferential family is introduced. F1, accuracy, precision and recall are on the 0–1 scale. Each population mean averages two seeds. Fixed test threshold 0.5; minority class is fitted from training labels. Sources: [13], accepted Tables 2–7,9 and 14. Primary adjusted effects remain in Table V95.1.

Table F1. v0.9.5. Selected-model descriptive test metrics for every task and policy.

Task / policy	Native BCE	Brier score	ECE 10 bins	Accuracy
checkerboard / fixed	0.0531 [0.0499, 0.0564]	0.0150 [0.0141, 0.0159]	0.0206 [0.0182, 0.0231]	0.9821 [0.9806, 0.9837]
checkerboard / halving	0.0491 [0.0482, 0.0501]	0.0139 [0.0135, 0.0142]	0.0190 [0.0178, 0.0203]	0.9833 [0.9821, 0.9845]
checkerboard / ETEL	0.0491 [0.0482, 0.0501]	0.0139 [0.0135, 0.0142]	0.0190 [0.0178, 0.0203]	0.9833 [0.9821, 0.9845]
spiral / fixed	0.0820 [0.0744, 0.0897]	0.0246 [0.0220, 0.0272]	0.0117 [0.0096, 0.0138]	0.9664 [0.9622, 0.9706]
spiral / halving	0.0789 [0.0701, 0.0877]	0.0237 [0.0208, 0.0266]	0.0099 [0.0077, 0.0121]	0.9672 [0.9628, 0.9716]
spiral / ETEL	0.0789 [0.0701, 0.0877]	0.0237 [0.0208, 0.0266]	0.0099 [0.0077, 0.0121]	0.9672 [0.9628, 0.9716]
annulus / fixed	0.3740 [0.3643, 0.3837]	0.1060 [0.1028, 0.1092]	0.0609 [0.0545, 0.0673]	0.8830 [0.8783, 0.8878]
annulus / halving	0.3708 [0.3623, 0.3792]	0.1051 [0.1021, 0.1081]	0.0592 [0.0523, 0.0661]	0.8835 [0.8786, 0.8884]
annulus / ETEL	0.3724 [0.3643, 0.3804]	0.1055 [0.1028, 0.1083]	0.0569 [0.0485, 0.0653]	0.8819 [0.8772, 0.8865]
disks / fixed	0.0105 [0.0086, 0.0124]	0.0031 [0.0024, 0.0038]	0.0044 [0.0038, 0.0051]	0.9957 [0.9945, 0.9969]
disks / halving	0.0098 [0.0074, 0.0121]	0.0030 [0.0022, 0.0038]	0.0038 [0.0033, 0.0043]	0.9957 [0.9946, 0.9969]
disks / ETEL	0.0098 [0.0074, 0.0121]	0.0030 [0.0022, 0.0038]	0.0038 [0.0033, 0.0043]	0.9957 [0.9946, 0.9969]

Mean and descriptive 95% interval from complete two-seed population means. No secondary inferential family or selection rule is added.

Table F2. v0.9.5. Selected-model descriptive test metrics for every task and policy.

Task / policy	Macro-F1 (0–1)	Balanced accuracy	Class 1 precision	Class 1 recall
checkerboard / fixed	0.9821 [0.9806, 0.9837]	0.9821 [0.9806, 0.9837]	0.9813 [0.9791, 0.9836]	0.9831 [0.9806, 0.9856]

Task / policy	Macro-F1 (0–1)	Balanced accuracy	Class 1 precision	Class 1 recall
checkerboard / halving	0.9833 [0.9821, 0.9845]	0.9833 [0.9821, 0.9845]	0.9833 [0.9820, 0.9847]	0.9834 [0.9808, 0.9860]
checkerboard / ETEL	0.9833 [0.9821, 0.9845]	0.9833 [0.9821, 0.9845]	0.9833 [0.9820, 0.9847]	0.9834 [0.9808, 0.9860]
spiral / fixed	0.9664 [0.9622, 0.9706]	0.9664 [0.9622, 0.9705]	0.9677 [0.9633, 0.9720]	0.9652 [0.9566, 0.9737]
spiral / halving	0.9672 [0.9628, 0.9716]	0.9672 [0.9628, 0.9716]	0.9677 [0.9623, 0.9732]	0.9668 [0.9590, 0.9746]
spiral / ETEL	0.9672 [0.9628, 0.9716]	0.9672 [0.9628, 0.9716]	0.9677 [0.9623, 0.9732]	0.9668 [0.9590, 0.9746]
annulus / fixed	0.8828 [0.8780, 0.8876]	0.8827 [0.8778, 0.8875]	0.8824 [0.8742, 0.8906]	0.8723 [0.8622, 0.8824]
annulus / halving	0.8833 [0.8783, 0.8882]	0.8832 [0.8782, 0.8882]	0.8818 [0.8719, 0.8917]	0.8742 [0.8631, 0.8853]
annulus / ETEL	0.8816 [0.8768, 0.8863]	0.8814 [0.8766, 0.8863]	0.8825 [0.8744, 0.8906]	0.8692 [0.8577, 0.8808]
disks / fixed	0.9894 [0.9865, 0.9923]	0.9893 [0.9854, 0.9932]	0.9817 [0.9746, 0.9889]	0.9810 [0.9731, 0.9889]
disks / halving	0.9895 [0.9867, 0.9922]	0.9896 [0.9855, 0.9937]	0.9811 [0.9737, 0.9886]	0.9817 [0.9733, 0.9901]
disks / ETEL	0.9895 [0.9867, 0.9922]	0.9896 [0.9855, 0.9937]	0.9811 [0.9737, 0.9886]	0.9817 [0.9733, 0.9901]

Mean and descriptive 95% interval from complete two-seed population means. No secondary inferential family or selection rule is added.

Table F3. v0.9.5. Selected-model descriptive test metrics for every task and policy.

Task / policy	Class 1 F1	Minority recall
checkerboard / fixed	0.9822 [0.9807, 0.9837]	0.9821 [0.9788, 0.9854]
checkerboard / halving	0.9834 [0.9821, 0.9846]	0.9834 [0.9810, 0.9858]
checkerboard / ETEL	0.9834 [0.9821, 0.9846]	0.9834 [0.9810, 0.9858]
spiral / fixed	0.9664 [0.9623, 0.9705]	0.9654 [0.9625, 0.9682]
spiral / halving	0.9672 [0.9630, 0.9715]	0.9650 [0.9608, 0.9692]
spiral / ETEL	0.9672 [0.9630, 0.9715]	0.9650 [0.9608, 0.9692]
annulus / fixed	0.8772 [0.8719, 0.8825]	0.8723 [0.8622, 0.8824]
annulus / halving	0.8779 [0.8723, 0.8834]	0.8742 [0.8631, 0.8853]
annulus / ETEL	0.8757 [0.8701, 0.8813]	0.8692 [0.8577, 0.8808]
disks / fixed	0.9813 [0.9761, 0.9865]	0.9810 [0.9731, 0.9889]

Task / policy	Class 1 F1	Minority recall
disks / halving	0.9813 [0.9765, 0.9862]	0.9817 [0.9733, 0.9901]
disks / ETEL	0.9813 [0.9765, 0.9862]	0.9817 [0.9733, 0.9901]

Mean and descriptive 95% interval from complete two-seed population means. No secondary inferential family or selection rule is added.

Table F0. v0.9.5 development-defined native validation-BCE targets used in Table F4.

Task	Frozen validation-native-BCE target
checkerboard	0.051100136185
spiral	0.061115478391
annulus	0.362627575541
disks	0.007526286300

Targets are displayed to 12 decimal places; full-precision values are retained in governance/MILESTONE_TARGETS_COSTREV01_v02.json, frozen by ROOT_MILESTONE_FREEZE_COSTREV01_02.json before confirmation. These development targets did not control scheduling. “Target reached” in Table F4 counts 16 portfolio-seed outcomes across eight independent populations; misses are censored, not assigned an imputed time. [13]

Table F4. v0.9.5. Actual policy cost and descriptive milestone coverage.

Task / policy	Charged CPU s [95% CI]	Mean native CPU s	Mean unused s	Mean overrun s	Target reached
checkerboard / fixed	1,449.0908 [1,362.3952, 1,535.7865]	1,449.09	75.61	0.00	5/16
checkerboard / halving	713.8184 [579.2639, 848.3728]	713.82	810.88	0.00	7/16
checkerboard / ETEL	1,463.0967 [1,381.3819, 1,544.8115]	1,463.10	61.60	0.00	7/16
spiral / fixed	1,087.6670 [1,087.3904, 1,087.9436]	1,087.67	2.95	0.00	0/16
spiral / halving	705.9600 [584.6060, 827.3139]	705.96	384.65	0.00	2/16
spiral / ETEL	1,087.6064 [1,087.0138, 1,088.1991]	1,087.61	3.01	0.00	2/16
annulus / fixed	1,155.7861 [1,155.3802, 1,156.1920]	1,155.79	3.13	0.00	11/16
annulus / halving	1,008.9102 [861.0910, 1,156.7293]	1,008.91	150.00	0.00	11/16
annulus / ETEL	1,157.3057 [1,156.6598, 1,157.9515]	1,157.31	1.61	0.00	11/16
disks / fixed	1,196.7744 [1,196.0953, 1,197.4536]	1,196.77	2.68	0.00	6/16

Task / policy	Charged CPU s [95% CI]	Mean native CPU s	Mean unused s	Mean overrun s	Target reached
disks / halving	737.9014 [615.9999, 859.8029]	737.90	461.55	0.00	8/16
disks / ETEL	1,197.0811 [1,196.4949, 1,197.6672]	1,197.08	2.37	0.00	8/16

Costs use population-first paired-seed means. Same ceiling does not mean same consumption. All attempts and closing tails remain charged.

Targets were frozen from development before confirmation and never control scheduling. Misses are censored; first-hit time is not imputed. Hit times and denominators are retained in evidence.

Table F5. v0.9.5. Late validation improvers and selected recipe counts.

Task / policy	Any miss /8	Mean largest validation gain	Selected recipes /16
checkerboard / fixed	4/8	0.0036	G121 base rate: 16
checkerboard / halving	0/8	0.0000	G121 base rate: 16
checkerboard / ETEL	0/8	0.0000	G121 base rate: 16
spiral / fixed	7/8	0.0019	G121 half rate: 1, G121 base rate: 15
spiral / halving	0/8	0.0000	G121 half rate: 2, G121 base rate: 14
spiral / ETEL	0/8	0.0000	G121 half rate: 2, G121 base rate: 14
annulus / fixed	5/8	0.0020	C_S base rate: 6, C_L base rate: 9, G121 half rate: 1
annulus / halving	5/8	0.0012	C_S base rate: 6, C_L base rate: 10
annulus / ETEL	4/8	0.0016	C_S base rate: 8, C_L base rate: 8
disks / fixed	8/8	0.0017	G121 base rate: 16
disks / halving	0/8	0.0000	G121 base rate: 16
disks / ETEL	0/8	0.0000	G121 base rate: 16

One preselected seed per population; six separate full 200-round references. Miss means a later eligible validation checkpoint improves on the original selected validation loss.

References are outside original policy budgets, not new independent replicates. These descriptive checks neither alter original selections nor establish test superiority.

Table F6. v0.9.5. Managed worker stages including retained failed attempts.

Stage	Jobs / attempts	Observed native CPU s	Charged CPU s	Summed lifetimes s
Engineering	39 / 51	8,743.75	8,758.90	8,945.94
Replacement development	24 / 24	26,769.77	26,769.77	27,316.29
Confirmation	192 / 192	207,375.97	207,375.97	211,274.76

Stage	Jobs / attempts	Observed native CPU s	Charged CPU s	Summed lifetimes s
Full references	192 / 192	48,801.53	48,801.53	49,838.14
Clean replays	4 / 4	3,012.42	3,012.42	3,090.23
Original development	24 / 24	21,788.97	21,788.97	22,298.24

Final engineering 39 includes initial 24; neither that subset nor checkpoint intervals are added twice. Original failed development remains charged and separate from the fresh replacement 24.

Summed overlapping process lifetimes are not project elapsed wall time. Six engineered crash charges retain conservative excess over the separately observed native CPU.

Table F7. v0.9.5. Official scoring whole-process lifetimes.

Operation	Native CPU s	Bootstrap CPU s	Native wall s	Bootstrap wall s
Author evaluation	1,976.08	0.03	2,131.74	2,131.88
Independent evaluation	3,235.86	0.00	3,302.90	3,303.03

No internal scoring timer is added to its owning native lifetime. Native and bootstrap wall lifetimes overlap within each operation and must not be added; the author and independent evaluation operations were sequential.

Failures and measurement scope

Engineering 39 jobs means 36 full trajectories (24 uninterrupted,6 planned pauses and 6 planned crashes) plus 3 short controller portfolios containing 18 candidate fixtures. Twelve retained interruption attempts bring the engineering record to 51 attempts; the initial 24 are already included. Recovery preserved complete optimizer, RNG, sampler, controller and counter state at committed boundaries. Four clean prescribed-action replays use fresh copied source/input directories with the preserved interpreter/libraries; they are not fresh installations or independent statistical replicates.

The original development 20-success/4-failure cohort was not scientifically accepted; all charges were retained before the reviewed replacement 24. Other retained failures include loader digest and attempt-inventory audit conventions, external interpreter confinement metadata, an F1 qualification denominator corrected before official evaluation, launcher OrderedDictionary inspection, resource observation and a nested reference-endpoint metadata assumption. Corrections were reviewed and versioned. Numerical checks were not silently rerun to erase an outer exit 1; missing auxiliary CPU remains unavailable rather than zero.

Author and independent official scoring checked 384 models under the frozen release. Maximum forward discrepancies were 1.94e−14 in probability,8.53e−14 in logit,1.78e−15 in gate weights and 9.24e−14 in expert values, with no discrete boundary disagreement. Numerical tolerance agreement is not bit identity. Retrospective module-location and copied-input checks have narrower scope than independent historical import observation. Whole native lifetimes include internal timers; nested round costs are not added again. Pure controller/serialization CPU was not isolated, and overlapping native/bootstrap or worker walls are not summed as project elapsed time.

The v0.9.5 evidence does not include new real-data confirmation, language-model training, physical-QPU execution, conversion or inference timing. All task/policy metrics and retained failures remain in the 27-page report and full evidence package. The defensible conclusion is recoverable, auditable execution without a demonstrated practical ETEL quality or training-efficiency advantage for this frozen pool and these tasks.